

EM Prelim Notes — Transcribed Problems

transcribed from `em_prelim_notes.pdf` · July 2, 2026

Transcribed page-by-page from the scanned handwritten notebook `em_prelim_notes.pdf` (115 pages, no text layer) on 2026-07-02. 56 entries in notebook order; the (*pp. N-M*) citation on each heading gives the PDF page numbers.

Conventions. $\mathfrak{r}$ (Fraktur r) stands for the notebook's Griffiths script-r separation vector, $\vec{\mathfrak{r}} = \vec{r} - \vec{r}'$, $\mathfrak{r} = |\vec{r} - \vec{r}'|$ (KaTeX has no lowercase script glyph). The notes' $\forall$ -like symbol for volume is kept as written. Bracketed italic lines describe the hand diagrams. [*sic: ...*] marks slips present in the notebook itself, transcribed as written. Entries tagged [*reference notes, not a worked problem*] are formula/summary sheets, not worked problems.

Duplicate scans in the PDF: pp. 26/27 are the same sheet scanned twice, p. 29 is a faint pencil duplicate of p. 30, and p. 112 duplicates p. 111 (different orientation).

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P1. E of Wire

(p. 1)

Electric field at a point a height z above the midpoint of a straight wire of length $2L$ (linear charge density λ) lying along $\hat{x}$; the Coulomb integral is set up and reduced to its vertical component.

Solution.

[Diagram: field point at height z above the midpoint of a wire of length $2L$ along $\hat{x}$; θ is the angle at the field point between the vertical and the separation vector $\mathbf{r}$ to a source point on the wire. A small side triangle identifies the legs: $r \rightarrow z$ (vertical), $r' \rightarrow x$ (horizontal), hypotenuse $\mathbf{r}$.]

$$\mathbf{r} = \sqrt{z^2 + x^2} \quad \vec{\mathbf{r}} = z \hat{z} - x \hat{x}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{\mathbf{r}}) \hat{\mathbf{r}}}{r^2} d\tau$$

$$\rho d\tau \rightarrow \lambda dx'$$

Horizontal components cancel.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda z \hat{z}}{(z^2 + x^2)^{3/2}} dx'$$

[The page ends here with the integral left unevaluated.]

P2. [Untitled — E on the Axis of a Charged Ring]

(p. 2)

Electric field at a point a height z above the center of a circular loop of radius r carrying uniform line charge λ .

Solution.

[Diagram: point P above the center of a circular loop; z runs from P down to the center, τ slants from P to the rim, with angle θ between them at P .]

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda \hat{\tau}}{\tau^2} dl'$$

Horizontal components cancel so then

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \int \frac{\cos \theta}{z^2 + r^2} dl$$

[Side triangle: legs z and r , hypotenuse τ , giving $\cos \theta = \frac{z}{\tau}$.]

$$\int dl = C = 2\pi r$$

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \frac{2\pi r z}{(z^2 + r^2)^{3/2}} \hat{z}$$

P3. B of Wire

(p. 3)

Magnetic field at point P a height z above the midpoint of a straight wire of length $2L$ along $\hat{x}$ carrying current I , by the Biot-Savart law.

Solution.

[Diagram: wire from $-L$ to L along $\hat{x}$ carrying current I ; point P above the midpoint, with $\vec{r}$ vertical to P, $\vec{r}'$ along the wire, $\mathfrak{r}$ from source point to P, angle θ at P.]

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \hat{\mathfrak{r}}}{\mathfrak{r}^2}$$

$$\vec{r} = z \hat{z} \quad \vec{r}' = x' \hat{x} \quad \vec{\mathfrak{r}} = \vec{r} - \vec{r}' = z \hat{z} - x' \hat{x}$$

$$d\vec{l} = dx' \hat{x} \quad \hat{\mathfrak{r}} = \frac{\vec{\mathfrak{r}}}{\mathfrak{r}} \quad \mathfrak{r}^2 = x^2 + z^2$$

$$d\vec{l} \times \vec{\mathfrak{r}} = (dx' \hat{x}) \times (z \hat{z} - x' \hat{x})$$

Horizontal components cancel; $\hat{x} \times \hat{x} = 0$

$$d\vec{l} \times \vec{\mathfrak{r}} = z dx' (\hat{x} \times \hat{z}) = z dx' (-\hat{y})$$

$$\vec{B} = -\frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{z dx'}{(x^2 + z^2)^{3/2}} \hat{y}$$

P4. B of Square Loop

(p. 4)

Magnetic field at a point a height z above the center of a square loop of side $2L$ carrying current I , assembled from the finite-wire result of the previous page.

Solution.

[Diagrams: (i) square loop of side $2L$ with apex point P above its center — s runs from P to the midpoint of one side, τ from P to a point on that side; (ii) triangle with vertical s , horizontal x , hypotenuse τ , angle θ ; (iii) triangle with vertical z , horizontal L , hypotenuse s , angle ψ . A corner sketch is scribbled out.]

$$\tau^2 = s^2 + x^2 \quad \cos \theta = \frac{s}{\tau} \quad \sin \psi = \frac{L}{s}$$

$$|\vec{B}_{\text{side}}| = -\frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{z dx'}{(x^2 + z^2)^{3/2}} \hat{y}$$

where for this problem $z \rightarrow s$ and all horizontal components cancel. For all four sides multiply by $4 \sin \psi$ so then

$$\vec{B}_{\text{tot}} = -\frac{\mu_0 I}{\pi} \int_{-L}^L \frac{s dx'}{(x^2 + s^2)^{3/2}} \frac{L}{s} \hat{z}$$

$$\vec{B}_{\text{tot}} = -\frac{\mu_0 I L}{\pi} \int_{-L}^L \frac{dx'}{(x^2 + s^2)^{3/2}} \hat{z}$$

P5. Charge Conservation

(p. 5)

Derives the continuity equation by taking the divergence of the Ampère–Maxwell law.

Solution.

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\underbrace{\nabla \cdot (\nabla \times \vec{B})}_{\text{zero}} = \mu_0 (\nabla \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \vec{E})$$

$$0 = (\nabla \cdot \vec{J}) \mu_0 + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \vec{E})$$

$$\nabla \cdot \vec{J} = -\epsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\epsilon_0} \right)$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

[The last line is boxed.]

P6. Poynting Theorem

(pp. 6-7)

Derives Poynting's theorem — the energy balance dW/dt for the electromagnetic field — starting from the work done by the Lorentz force on a charge distribution, and the local statement $\partial u/\partial t = -\nabla \cdot \vec{S}$ when no work is done.

Solution.

$$\vec{F} \cdot d\vec{l} = q \left(\vec{E} + \vec{v} \times \vec{B} \right) \cdot \vec{v} dt = q \vec{E} \cdot \vec{v} dt$$

where $q \rightarrow \rho d\tau$

$$\vec{F} \cdot d\vec{l} = q \vec{E} \cdot \vec{v} dt \rightarrow \vec{E} \cdot \rho \vec{v} dt d\tau$$

where $\rho \vec{v} \rightarrow \vec{J}$

$$\vec{F} \cdot d\vec{l} = \vec{E} \cdot \vec{J} d\tau dt$$

where $dW = \vec{F} \cdot d\vec{l}$, so

$$\frac{dW}{dt} = \int_V \left(\vec{E} \cdot \vec{J} \right) d\tau$$

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \vec{E} \cdot \left(\nabla \times \vec{B} \right) - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

prod. rule 6 [*margin note: "pg 23 Camb 2.53"*]

$$\nabla \cdot \left(\vec{E} \times \vec{B} \right) = \vec{B} \cdot \left(\nabla \times \vec{E} \right) - \vec{E} \cdot \left(\nabla \times \vec{B} \right)$$

Faraday's law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{E} \cdot \left(\nabla \times \vec{B} \right) = -\vec{B} \cdot \frac{\partial \vec{E}}{\partial t} - \nabla \cdot \left(\vec{E} \times \vec{B} \right)$$

where

$$\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (B^2) \quad \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (E^2)$$

so

$$\vec{E} \cdot \vec{J} = -\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B})$$

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint_{SA} (\vec{E} \times \vec{B}) \cdot d\vec{a}$$

[The line above is boxed.]

$$\vec{S} \equiv \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V u d\tau - \oint_S \vec{S} \cdot d\vec{a}$$

[The line above is boxed.]

if no work is done

$$\int \frac{\partial u}{\partial t} d\tau = - \oint \vec{S} \cdot d\vec{a} = - \int (\nabla \cdot \vec{S}) d\tau$$

$$\frac{\partial u}{\partial t} = -\nabla \cdot \vec{S}$$

P7. [Untitled — Line Charge above a Grounded Plane (Method of Images)]

(pp. 8-9)

An infinite line charge λ at height z_0 above the grounded plane $z = 0$ ($V = 0$): the field of a wire from Gauss's law, its logarithmic potential, and the total potential above the plane by the method of images.

Solution.

[Diagram: $\hat{z}$ axis vertical; a line charge λ drawn horizontally through the axis at height z_0 ; below it the x - y plane marked $V = 0$, with axes $\hat{y}$ (right) and $\hat{x}$ (toward viewer).]

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \rightarrow E(2\pi RL) = \frac{\lambda L}{\epsilon_0}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 R}$$

$$V = - \int \vec{E} \cdot d\vec{l} = - \int_a^s \frac{\lambda}{2\pi\epsilon_0 r} dr$$

$$V = - \frac{\lambda}{2\pi\epsilon_0} (\ln(s) - \ln(a)) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{s}\right)$$

using method of images

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{r}\right)$$

so

$$V_1 = \frac{\lambda}{2\pi\epsilon_0} \ln \left[\frac{a}{\left(x^2 + y^2 + (z - z_0)^2\right)^{1/2}} \right]$$

[sic: for an infinite line the distance should involve only the coordinates transverse to the wire; the notes fold in the full $x^2 + y^2$, defining $s^2 = x^2 + y^2$ below.]

$$V_2 = \frac{\lambda'}{2\pi\epsilon_0} \ln \left[\frac{a}{\left(x^2 + y^2 + (z + z_0)^2\right)^{1/2}} \right]$$

$$V_{\text{tot}} = \frac{\lambda}{2\pi\epsilon_0} \left\{ \ln \left[\frac{a}{\left(s^2 + (z - z_0)^2\right)^{1/2}} \right] - \ln \left[\frac{a}{\left(s^2 + (z + z_0)^2\right)^{1/2}} \right] \right\} \quad (s^2 = x^2 + y^2)$$

@ $V = 0$

$$\therefore V_{\text{tot}} = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{s^2 + (z - z_0)^2}{s^2 + (z + z_0)^2} \right]$$

[The line above is boxed. sic: the ratio is inverted relative to the preceding line, which combines to $\ln[(s^2 + (z + z_0)^2)/(s^2 + (z - z_0)^2)]$.]

P8. B of Loop & Rotating Spherical Shell

(p. 10)

On-axis magnetic field of a circular current loop of radius R by Biot-Savart, then — treating a rotating charged spherical shell (radius R , surface density σ , angular velocity ω) as a stack of loops — the field at its center.

Solution.

[Diagram: point P at the apex, dashed height z down to the center of a circle of radius R , slant $\mathfrak{r}$ from P to the rim.]

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{d\vec{I} \times \hat{\mathfrak{r}}}{\mathfrak{r}^2} \quad \vec{\mathfrak{r}} = z \hat{z} - R \hat{r} \quad \mathfrak{r}^2 = R^2 + z^2$$

$$d\vec{l} \times \vec{\mathfrak{r}} = R d\phi \hat{\phi} \times (z \hat{z} - R \hat{r})$$

due to symmetry $\hat{\phi} \times \hat{z} = \hat{r}$, $\hat{\phi} \times (-\hat{r}) = \hat{z}$; $\hat{r}$, $\hat{\phi}$ components cancel, leaving

$$d\vec{l} \times \vec{\mathfrak{r}} = R^2 d\phi \hat{z} \quad \frac{\hat{\mathfrak{r}}}{\mathfrak{r}^2} = \frac{\vec{\mathfrak{r}}}{\mathfrak{r}^3} \quad d\vec{I} = I d\vec{l}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{R^2 d\phi}{(R^2 + z^2)^{3/2}} \hat{z} \quad \text{let } R \equiv R'$$

for 3D shell $\rightarrow \vec{K} = \sigma \vec{v}$ where $\vec{v} = R' \omega$

$$R' \rightarrow R \sin \theta \quad z \rightarrow R \cos \theta \quad \& \quad I \equiv R' \sigma \omega$$

$$I \rightarrow \sigma R \sin \theta \omega \quad d\vec{l} = R d\theta \quad d\vec{I} = \sigma R^2 \sin \theta \omega d\theta$$

$$\vec{B} = \frac{\mu_0}{2} \int \frac{R^2 \sin^2 \theta dI}{\left(R^2 (\sin^2 \theta + \cos^2 \theta)\right)^{3/2}} \hat{z} = \frac{\mu_0 R \sigma \omega}{2} \int \sin^3 \theta d\theta \hat{z}$$

[The page ends here with the $\sin^3 \theta$ integral unevaluated.]

P9. B outside Spinning Solid Sphere

(pp. 11-14)

A uniformly charged solid sphere (radius R , total charge Q , density ρ) spins at angular velocity ω : the vector potential is built by superposing spinning-shell dipole contributions split at the field radius r , and then $\vec{B} = \nabla \times \vec{A}$ is evaluated in spherical coordinates.

Solution.

$$\vec{A}_{\text{dip}} \approx \frac{\mu_0}{4\pi r^2} (\vec{m} \times \hat{r}) \quad \sigma \equiv \rho dr' \quad V = \frac{4}{3}\pi R^3 \quad \rho = \frac{Q}{V}$$

$$\vec{K} = \sigma \vec{v} = \sigma \omega r' \hat{z} \rightarrow d\vec{J} = \rho \omega r' dr' \hat{z}$$

[sic: $\vec{K}$ and $d\vec{J}$ circulate along $\hat{\phi}$; the $\hat{z}$ written here is the axis that reappears in $d\vec{m} \parallel \hat{z}$.]

$$d\vec{m} = V d\vec{J} = \frac{4}{3}\pi R^3 \rho \omega r' dr' \hat{z}$$

$$\hat{z} \times \hat{r} = \hat{\phi}$$

however for spherical

$$d\phi \hat{\phi} \rightarrow \sin \theta d\phi \hat{\phi}$$

$$\vec{A} \approx \frac{\mu_0}{4\pi r^2} \int \frac{4}{3}\pi R^3 \rho \omega r' dr' \sin \theta \hat{\phi}$$

$$\vec{A} \approx \frac{\mu_0 \rho \omega \sin \theta}{3} \int \frac{R^3 r' dr'}{r^2} \hat{\phi}$$

$$\vec{A} = \frac{\mu_0 \rho \omega \sin \theta}{3} \int_0^r \frac{R_{<}^3 r' dr'}{r^2} \hat{\phi} + \frac{\mu_0 \rho \omega \sin \theta}{3} \int_r^R \frac{R_{>}^3 r' dr'}{r^2} \hat{\phi} \quad (\text{let } R_{<} \equiv r', \ R_{>} \equiv r)$$

$$\vec{A} = \frac{\mu_0 \rho \omega \sin \theta}{3} \left[\int_0^r \frac{r'^4}{r^2} dr' + \int_r^R r r' dr' \right] \hat{\phi}$$

$$\rightarrow \frac{\mu_0 \rho \omega \sin \theta}{3} \left[\frac{r^5}{5r^2} + \frac{r}{2} (R^2 - r^2) \right] \hat{\phi}$$

$$\rightarrow \frac{\mu_0 \rho \omega \sin \theta}{3} \left[\frac{r^3}{5} + \frac{R^2 r}{2} - \frac{r^3}{2} \right] \hat{\phi}$$

$$\rightarrow \frac{\mu_0 \rho \omega r \sin \theta}{3} \left[\frac{r^2}{5} - \frac{r^2}{2} + \frac{R^2}{2} \right] \hat{\phi}$$

$$\rightarrow \frac{\mu_0 \rho \omega r \sin \theta}{2} \left[\frac{R^2}{3} - \frac{r^2}{5} \right] \hat{\phi}$$

$$\vec{A} = \frac{\mu_0 \rho \omega r \sin \theta}{2} \left[\frac{R^2}{3} - \frac{r^2}{5} \right] \hat{\phi}$$

[The line above is boxed.]

$$\vec{B} = \nabla \times \vec{A} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial r} & \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} & \frac{1}{r} \frac{\partial}{\partial \phi} \\ 0 & 0 & r A_\phi \sin \theta \end{vmatrix}$$

$$\vec{B} = \frac{\hat{r}}{r^2 \sin \theta} \partial_\theta (r A_\phi \sin \theta) - \frac{\hat{\theta}}{r \sin \theta} \partial_r (r A_\phi \sin \theta)$$

$$= \frac{1}{r^2 \sin \theta} \partial_\theta \left[\frac{\mu_0 \rho \omega r^2 \sin^2 \theta}{2} \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \right] \hat{r} - \frac{1}{r \sin \theta} \partial_r \left[\frac{\mu_0 \rho \omega r^2 \sin^2 \theta}{2} \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \right] \hat{\theta}$$

$$= \frac{2 \sin \theta \cos \theta}{\sin \theta} \frac{\mu_0 \rho \omega}{2} \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \hat{r} - \frac{1}{r \sin \theta} \frac{\mu_0 \rho \omega \sin^2 \theta}{2} \partial_r \left(\frac{r^2 R^2}{3} - \frac{r^4}{5} \right) \hat{\theta}$$

$$\vec{B} = \mu_0 \rho \omega \cos \theta \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \hat{r} - \frac{\mu_0 \rho \omega \sin \theta}{2r} \left(\frac{2r R^2}{3} - \frac{4r^3}{5} \right) \hat{\theta}$$

$$\vec{B} = \mu_0 \rho \omega \left[\left(\frac{R^2}{3} - \frac{r^2}{5} \right) \cos \theta \hat{r} - \left(\frac{R^2}{3} - \frac{2r^2}{5} \right) \sin \theta \hat{\theta} \right]$$

$$\rho = \frac{Q}{\frac{4}{3} \pi R^3}$$

$$\vec{B}_{\text{out}} = \frac{\mu_0 \omega Q}{4\pi R} \left[\left(1 - \frac{3r^2}{5R^2} \right) \cos \theta \hat{r} - \left(1 - \frac{6r^2}{5R^2} \right) \sin \theta \hat{\theta} \right]$$

[The line above is boxed. sic: despite the title and the "out" subscript, this is the interior field ($r \leq R$) — it follows from the $r' < r$ / $r' > r$ shell split above.]

P10. Metal Sphere with Linear Dielectric (Ex 4.5)

(pp. 15-16)

A metal sphere of radius a carrying free charge Q is surrounded by a linear dielectric shell (permittivity ϵ , susceptibility χ_e) out to radius b : find $\vec{D}$, $\vec{E}$, the potential at the origin, and the bound charges. [The margin tag "Ex 4.5" is struck through; pages carry corner marks 1/2, 2/2.]

Solution.

[Diagram: concentric circles — inner sphere of radius a with charge Q , dielectric shell out to radius b .]

$$\oint_{SA} \vec{D} \cdot d\vec{a} = Q_{f,\text{enc}} \quad SA = 4\pi r^2 \quad Q_{f,\text{enc}} = Q$$

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r} \quad (r > a)$$

Inside metal sphere $\vec{E} = \vec{P} = \vec{D} = 0$

$$\vec{E} = \begin{cases} \frac{Q}{4\pi\epsilon r^2} \hat{r} & a < r < b \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} & r > b \end{cases}$$

$$V = - \int_{\infty}^0 \vec{E} \cdot d\vec{l} = - \int_{\infty}^b E_{\text{out}} dr - \int_b^a E_{\text{mat}} dr - \int_a^0 E_{\text{in}} dr$$

$$V = \frac{-Q}{4\pi} \left[\int_{\infty}^b \frac{1}{\epsilon_0 r^2} dr + \int_b^a \frac{1}{\epsilon r^2} dr \right]$$

[Underbraces mark the two integrals as $-\frac{1}{\epsilon_0} \left(\frac{1}{b} - \frac{1}{\infty} \right)$ and $-\frac{1}{\epsilon} \left(\frac{1}{a} - \frac{1}{b} \right)$.]

$$V_{\text{org}} = \frac{Q}{4\pi} \left[\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b} \right]$$

[The line above is boxed.]

$$\vec{P} = \epsilon_0 \chi_e \vec{E} = \epsilon_0 \chi_e \frac{Q}{4\pi\epsilon r^2} \hat{r}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = 0$$

while

$$\sigma_b = \vec{P} \cdot \hat{n} = \begin{cases} \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon b^2} & \text{outer surface} \\ \frac{-\epsilon_0 \chi_e Q}{4\pi \epsilon a^2} & \text{inner surface} \end{cases}$$

P11. Spherical Conductor w/ Linear Dielectric (Prob 4.26)

(pp. 17-18)

Energy of the configuration: a spherical conductor of inner radius a with charge Q , surrounded out to radius b by linear dielectric material χ_e ; computes $W = \frac{1}{2} \int \vec{D} \cdot \vec{E} d\tau$.

Solution.

Inner radius a w/ charge Q . Outer radius b w/ linear dielectric material χ_e .

[Diagram: sphere of radius a carrying Q , surrounded by a dielectric layer out to radius b .]

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r} \quad (r > a)$$

$$\vec{E} = \begin{cases} \frac{Q}{4\pi\epsilon r^2} \hat{r} & (a < r < b) \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} & (r > b) \end{cases}$$

$$u = \frac{1}{2} (\vec{D} \cdot \vec{E} + \vec{B} \cdot \vec{H}) \quad W = \int u d\tau$$

[The $\vec{B} \cdot \vec{H}$ term is struck through with a 0 above it.]

$$W = \frac{1}{2} \int \vec{D} \cdot \vec{E} d\tau = \frac{1}{2} \left[\int_a^b (\vec{D} \cdot \vec{E}) d\tau + \int_b^\infty (\vec{D} \cdot \vec{E}) d\tau \right]$$

$$= \frac{1}{2} \frac{Q^2}{(4\pi)^2} \left[\int_a^b \frac{1}{\epsilon r^4} d\tau + \int_b^\infty \frac{1}{\epsilon_0 r^4} d\tau \right]$$

$$\int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta = 4\pi$$

$$W = \frac{4\pi}{2} \frac{Q^2}{(4\pi)^2} \left[\frac{1}{\epsilon} \int_a^b \frac{r^2}{r^4} dr + \frac{1}{\epsilon_0} \int_b^\infty \frac{r^2}{r^4} dr \right]$$

$$\epsilon_r = 1 + \chi_e \quad \epsilon_0 \epsilon_r = \epsilon$$

$$W = \frac{Q^2}{8\pi\epsilon_0} \left\{ \frac{1}{(1 + \chi_e)} \int_a^b r^{-2} dr + \int_b^\infty r^{-2} dr \right\}$$

[Underbraces mark the integrals as $-r^{-1}|_a^b$ and $-r^{-1}|_b^\infty$.]

$$W = \frac{Q^2}{8\pi\epsilon_0} \left[\frac{\left(\frac{1}{a} - \frac{1}{b}\right)}{(1 + \chi_e)} + \frac{1}{b} \right]$$

$$W = \frac{Q^2}{8\pi\epsilon_0 (1 + \chi_e)} \left[\frac{1}{a} + \frac{\chi_e}{b} \right]$$

[The line above is boxed.]

P12. Loop Inductance (Prob 7.22)

(p. 19)

A small loop of radius a sits on the axis of a large loop of radius b , a height z above it: the flux through the little loop from current I in the big loop, and the mutual-inductance reciprocity $M_{12} = M_{21}$ checked via the little loop's dipole field.

Solution.

[Diagram: small loop of radius a at height z above a coaxial large loop of radius b ; side triangle with legs z, b and hypotenuse r .]

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \hat{r}}{r^2}$$

$$dl = b d\phi \hat{\phi} \quad \vec{r} = z \hat{z} - b \hat{r} \quad r^2 = b^2 + z^2 \quad d\vec{l} \times \vec{r} = b^2 d\phi \hat{z}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{b^2 d\phi \hat{z}}{(b^2 + z^2)^{3/2}}$$

$$\vec{B} = \frac{\mu_0 I}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \hat{z}$$

$$\Phi = \int \vec{B} \cdot d\vec{a} = B \pi a^2 = \frac{\mu_0 I \pi a^2 b^2}{2 (b^2 + z^2)^{3/2}} = \Phi_a$$

[The flux result is boxed.]

$$\Phi_1 = M_{12} I_2 \quad \Phi_2 = M_{21} I_1$$

$$M_{12} = M_{21}$$

$$\vec{B}_b = \vec{B}_{\text{dip}} \quad |\vec{m}_a| = I \pi a^2$$

where $r = \sqrt{b^2 + z^2}$, $\sin \theta = \frac{b}{r}$ — gives same flux.

P13. ||-Plate Cap. w/ 2 Linear Dielectric (Problem 4.18)

(pp. 20-21)

A parallel-plate capacitor with plate charge $\pm\sigma$ is filled by two linear dielectric slabs, each of thickness a : slab 1 ($\epsilon_1 = 2\epsilon_0$) against the positive plate, slab 2 ($\epsilon_2 = \frac{3}{2}\epsilon_0$) below it. Find D , E , P , the potential difference, the bound charges, and re-check the fields from all the surface charges.

Solution.

[Diagram: 3-D sketch of the stack — hatched top plate at $+\sigma$, slab 1 then slab 2 (thicknesses a and a), bottom plate at $-\sigma$; $\hat{z}$ drawn pointing downward.]

$$\epsilon_1 = 2\epsilon_0 \quad \epsilon_2 = \frac{3}{2}\epsilon_0$$

$$\oint \vec{D} \cdot d\vec{a} = Q_{f,enc} \quad DA = \sigma A$$

$$D = \sigma \quad \hat{z}$$

[The line above is boxed.]

$$\vec{D} = \epsilon \vec{E} \rightarrow \vec{E} = \frac{\sigma}{\epsilon_1} \text{ slab 1} \quad \vec{E} = \frac{\sigma}{\epsilon_2} \text{ slab 2}$$

$$E_1 = \frac{\sigma}{2\epsilon_0} \quad E_2 = \frac{2\sigma}{3\epsilon_0} \quad \hat{z}$$

[The line above is boxed.]

$$\vec{P} = \epsilon_0 \chi_e \vec{E} \quad \epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0} \quad \epsilon_r - 1 = \chi_e$$

$$\vec{P} = \epsilon_0 (\epsilon_r - 1) \vec{E}$$

$$P_1 = \epsilon_0 (2 - 1) E_1 = \frac{\epsilon_0 \sigma}{2\epsilon_0} \quad P_1 = \frac{\sigma}{2}$$

$$P_2 = \epsilon_0 \left(\frac{3}{2} - 1 \right) E_2 = \frac{\epsilon_0}{2} \frac{2\sigma}{3\epsilon_0} \quad P_2 = \frac{\sigma}{3}$$

[The $P_1 = \sigma/2$ and $P_2 = \sigma/3$ results are boxed.]

$$V = E_1 a + E_2 a = \frac{\sigma a}{2\epsilon_0} + \frac{2\sigma a}{3\epsilon_0}$$

$$V = \frac{\sigma a}{\epsilon_0} \left(\frac{1}{2} + \frac{2}{3} \right) = \frac{\sigma a}{\epsilon_0} \left(\frac{3+4}{6} \right) = \frac{7\sigma a}{6\epsilon_0}$$

$$V = \frac{7\sigma a}{6\epsilon_0}$$

[The line above is boxed.]

$$\rho_b = 0$$

[Diagram: the stack redrawn with the surface charges circled at each boundary — $+\sigma$ (top plate), $-\sigma/2$ and $+\sigma/2$ (top and bottom faces of slab 1), $-\sigma/3$ and $+\sigma/3$ (top and bottom faces of slab 2), $-\sigma$ (bottom plate).]

(slab 1)

$$\sum_{\sigma \text{ above}} = \sigma - \frac{\sigma}{2}$$

$$\sum_{\sigma \text{ below}} = \frac{\sigma}{2} - \frac{\sigma}{3} + \frac{\sigma}{3} - \sigma = -\frac{\sigma}{2}$$

$$\rightarrow E_1 = \frac{\sigma}{2\epsilon_0}$$

similarly for slab 2

$$\text{above: } \sigma - \frac{\sigma}{3} = \frac{2\sigma}{3} \quad \text{below: } \frac{\sigma}{3} - \sigma = -\frac{2\sigma}{3}$$

$$E_2 = \frac{2\sigma}{3\epsilon_0} \checkmark$$

P14. Multipole Expansion

(pp. 22-23)

Potential of a physical dipole: charges $+q$ and $-q$ separated by a distance d , evaluated at a field point a distance r from the center at angle θ from the dipole axis. First the far-field ($r \gg d$) binomial expansion giving the dipole potential, then the full Legendre-series expansion giving the odd multipoles (dipole, quadrupole $\rightarrow 0$, octopole). (The page carries a struck-through problem tag, apparently "Prob 3.3...", before the title.)

Solution.

[Diagram: $+q$ above $-q$ separated by d ; field point at the end of $\vec{r}$ drawn from the dipole center at angle θ from the axis, with separation vectors $\mathbf{r}_+$ from $+q$ and $\mathbf{r}_-$ from $-q$.]

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\mathbf{r}_+} - \frac{q}{\mathbf{r}_-} \right)$$

$$\mathbf{r}_{\pm}^2 = r^2 + \left(\frac{d}{2}\right)^2 \mp rd \cos \theta = r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)$$

For $r \gg d$ the 3rd term is negligible.

$$\frac{1}{\mathbf{r}_{\pm}} \approx \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$

$$\frac{1}{\mathbf{r}_+} - \frac{1}{\mathbf{r}_-} \approx \frac{d}{r^2} \cos \theta \longrightarrow V(\vec{r}) \approx \frac{1}{4\pi\epsilon_0} \frac{dq \cos \theta}{r^2}$$

[Dashed rule; the general expansion follows.]

$$\frac{1}{\mathbf{r}} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \alpha)$$

choosing $\mathbf{r} \rightarrow \mathbf{r}_+$, $r' \rightarrow \frac{d}{2}$, $\alpha \rightarrow \theta$

$$\frac{1}{\mathbf{r}_+} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n P_n(\cos \theta)$$

$\mathbf{r}_-$: $\theta \rightarrow \theta + 180 \longrightarrow \cos \theta \rightarrow -\cos \theta$

$$\frac{1}{\mathbf{r}_-} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n P_n(-\cos \theta)$$

$$P_n(-x) = (-1)^n P_n(x)$$

$$\begin{aligned}
V &= \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{\mathbf{r}_+} - \frac{1}{\mathbf{r}_-} \right) = \frac{1}{4\pi\epsilon_0} q \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n [P_n(\cos \theta) - P_n(-\cos \theta)] \\
&= \frac{2q}{4\pi\epsilon_0 r} \sum_{n=1,3,5,\dots} \left(\frac{d}{2r} \right)^n P_n(\cos \theta)
\end{aligned}$$

dipole ($n = 1$) $\longrightarrow$

$$V_{dip} = \frac{2q}{4\pi\epsilon_0 r} \frac{d}{2r} P_1(\cos \theta) = \frac{qd \cos \theta}{4\pi\epsilon_0 r^2}$$

Quadrupole ($n = 2$) $\longrightarrow$ zero. Octopole ($n = 3$):

$$V_{oct} = \frac{2q}{4\pi\epsilon_0} \frac{1}{r} \left(\frac{d}{2r} \right)^3 P_3(\cos \theta), \quad P_3(x) = \frac{5x^3 - 3x}{2}$$

$$V_{oct} = \frac{2q}{4\pi\epsilon_0} \frac{1}{r} \frac{d^3}{8r^3} \left(\frac{5 \cos^3 \theta - 3 \cos \theta}{2} \right)$$

$$V_{oct} = \frac{qd^3}{4\pi\epsilon_0} \frac{1}{8r^4} (5 \cos^3 \theta - 3 \cos \theta)$$

P15. Reflection & Transmission @ Oblique Incidence

(pp. 24-28)

A monochromatic plane wave in medium 1 strikes the interface with medium 2 at incidence angle θ_I , with the polarization in the plane of incidence. Phase matching at the boundary gives the three laws of geometrical optics; the four electromagnetic boundary conditions then give the Fresnel equations for $\vec{E}_{0R}$ and $\vec{E}_{0T}$. Four-page entry marked 1/4-4/4. [pp. 26 and 27 are duplicate scans of the same sheet (3/4).]

Solution.

$$\vec{E}_I(\vec{r}, t) = \tilde{E}_{0I} e^{i(\vec{k}_I \cdot \vec{r} - \omega t)}, \quad \vec{B}_I(\vec{r}, t) = \frac{1}{v_1} (\hat{k}_I \times \vec{E}_I)$$

similarly for $\vec{E}_R, \vec{B}_R, \vec{E}_T, \vec{B}_T$.

[Diagram: hatched interface with normal along $\hat{z}$, $\hat{x}$ along the surface; $\vec{k}_I$ incident at θ_I , $\vec{k}_R$ reflected at θ_R , $\vec{k}_T$ transmitted.]

$$k_I v_1 = k_R v_1 = k_T v_2 = \omega$$

$$k_I = k_R = \frac{v_2}{v_1} k_T = \frac{n_1}{n_2} k_T$$

$$(\) e^{i(\vec{k}_I \cdot \vec{r} - \omega t)} + (\) e^{i(\vec{k}_R \cdot \vec{r} - \omega t)} = (\) e^{i(\vec{k}_T \cdot \vec{r} - \omega t)} \quad (z = 0)$$

$$\vec{k}_I \cdot \vec{r} = \vec{k}_R \cdot \vec{r} = \vec{k}_T \cdot \vec{r} \quad (z = 0)$$

$$x(k_I)_x + y(k_I)_y = x(k_R)_x + y(k_R)_y = x(k_T)_x + y(k_T)_y$$

if $x = 0$:

$$(k_I)_y = (k_R)_y = (k_T)_y$$

if $y = 0$:

$$(k_I)_x = (k_R)_x = (k_T)_x$$

where $\vec{k}$ lies in the x-z plane, so $(k_I)_y = 0$ (1st Law — plane of incidence). So

$$k_I \sin \theta_I = k_R \sin \theta_R = k_T \sin \theta_T$$

$$\theta_I = \theta_R$$

(2nd Law — "angle of refraction") [*sic: $\theta_I = \theta_R$ is the law of reflection*]

$$k_I \sin \theta_I = k_T \sin \theta_T \quad \text{where } k_I = \frac{n_1}{n_2} k_T$$

so

$$\frac{\sin \theta_T}{\sin \theta_I} = \frac{n_1}{n_2} \quad (3\text{rd Law — Snell's})$$

B.C.'s:

$$(i) \quad \epsilon_1 \left(\vec{E}_{0I} + \vec{E}_{0R} \right)_z = \epsilon_2 \left(\vec{E}_{0T} \right)_z$$

$$(ii) \quad \left(\vec{B}_{0I} + \vec{B}_{0R} \right)_z = \left(\vec{B}_{0T} \right)_z$$

$$(iii) \quad \left(\vec{E}_{0I} + \vec{E}_{0R} \right)_{x,y} = \left(\vec{E}_{0T} \right)_{x,y}$$

$$(iv) \quad \frac{1}{\mu_1} \left(\vec{B}_{0I} + \vec{B}_{0R} \right)_{x,y} = \frac{1}{\mu_2} \left(\vec{B}_{0T} \right)_{x,y}$$

[Diagram: media ① and ② separated by the hatched interface; $\vec{E}_I, \vec{E}_R, \vec{E}_T$ drawn in the plane of incidence with $\vec{B}$'s perpendicular to their $\vec{k}$'s; angles $\theta_I, \theta_R, \theta_T$ measured from the z-axis.]

$$(i) \quad \epsilon_1 \left(-\tilde{E}_{0I} \sin \theta_I + \tilde{E}_{0R} \sin \theta_R \right) = \epsilon_2 \left(-\tilde{E}_{0T} \sin \theta_T \right) \quad (1)$$

$\vec{B} = \frac{1}{v_1}(\hat{k} \times \vec{E})$; (ii) does nothing ($B_z = 0$).

$$(iii) \quad \tilde{E}_{0I} \cos \theta_I + \tilde{E}_{0R} \cos \theta_R = \tilde{E}_{0T} \cos \theta_T \quad (2)$$

$$(iv) \quad \frac{1}{\mu_1} \left(\tilde{B}_{0I} + \tilde{B}_{0R} \right)_{x,y} = \frac{1}{\mu_2} \left(\tilde{B}_{0T} \right)_{x,y}$$

$$\vec{B}_0 = \frac{1}{v_1} \left(\hat{k} \times \vec{E}_0 \right)$$

$$\frac{1}{\mu_1 v_1} \left(\tilde{E}_{0I} - \tilde{E}_{0R} \right) = \frac{1}{\mu_2 v_2} \tilde{E}_{0T} \quad (3)$$

$$\longrightarrow \quad \tilde{E}_{0_I} - \tilde{E}_{0_R} = \frac{\mu_1 v_1}{\mu_2 v_2} \tilde{E}_{0_T}$$

let

$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}, \quad \tilde{E}_{0_I} - \tilde{E}_{0_R} = \beta \tilde{E}_{0_T} \quad (4)$$

$\theta_I = \theta_R$, so

$$(2) \longrightarrow \quad \tilde{E}_{0_I} + \tilde{E}_{0_R} = \frac{\cos \theta_T}{\cos \theta_I} \tilde{E}_{0_T}$$

let $\alpha \equiv \frac{\cos \theta_T}{\cos \theta_I}$, so

$$\tilde{E}_{0_I} + \tilde{E}_{0_R} = \alpha \tilde{E}_{0_T} \quad (5)$$

(4) + (5) $\longrightarrow$

$$2\tilde{E}_{0_I} = (\alpha + \beta) \tilde{E}_{0_T} \quad (6)$$

(5) - (4) $\longrightarrow$

$$2\tilde{E}_{0_R} = (\alpha - \beta) \tilde{E}_{0_T} \quad \longrightarrow \quad \frac{2\tilde{E}_{0_R}}{\alpha - \beta} = \tilde{E}_{0_T}$$

$$(6) \longrightarrow \quad \tilde{E}_{0_I} = \left(\frac{\alpha + \beta}{\alpha - \beta} \right) \tilde{E}_{0_R}$$

(the 2's cancel), so

$$\tilde{E}_{0_T} = \left(\frac{2}{\alpha + \beta} \right) \tilde{E}_{0_I}$$

$$\tilde{E}_{0_R} = \left(\frac{\alpha - \beta}{\alpha + \beta} \right) \tilde{E}_{0_I}$$

(both boxed) — Fresnel's equations.

P16. Ex 5.1 Cyclotron Motion

(pp. 29-30)

Griffiths Ex 5.1: a charge Q moves counterclockwise with speed v around a circle of radius R in the uniform field $\vec{B} = -B\hat{z}$; the magnetic force supplies the centripetal acceleration, giving the cyclotron momentum formula. *[p. 29 is a faint pencil-scan duplicate of the same sheet as p. 30.]*

Solution.

$$\vec{B} = -B\hat{z}$$

Q moves counterclockwise w/ speed v @ radius R .

$$F_{mag} = QvB = ma, \quad a = \frac{v^2}{R}$$

$$QvB = m \frac{v^2}{R} \longrightarrow p = QBR$$

$$QBR = mv \longrightarrow p = mv$$

[Diagram: circular orbit of radius R in the x - y plane, with the charge Q on the circle carrying tangential $\vec{v}$ and inward $\vec{F}$, axes x, y, z drawn. Beside it: a helix wound along the field direction, with arrows B along the axis and $v_{\parallel}$ parallel to it — the trajectory when the particle also has a velocity component along $\vec{B}$.]

P17. B-field on Atomic Orbits

(pp. 31-32)

Effect of a magnetic field on an electron's circular atomic orbit (radius R , speed v): the orbit is treated as a current loop with orbital dipole moment $\vec{m}$; switching on $\vec{B}$ adds the force $-e(\vec{v} \times \vec{B})$, speeding the electron to v' at fixed R , and the resulting $\Delta\vec{m}$ is opposite to $\vec{B}$ (diamagnetism).

Solution.

$$T = \frac{2\pi R}{v} \quad (\text{Period})$$

$$I = \frac{-e}{T} = \frac{-ev}{2\pi R}$$

Orbital dipole moment ($I\pi R^2$):

$$\vec{m} = \frac{-ev}{2\pi R} \pi R^2 = -\frac{1}{2} evR \hat{z}$$

$$\vec{F} = m\vec{a}$$

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = m_e \frac{v^2}{R}$$

w/ additional force $-e(\vec{v} \times \vec{B})$:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} + ev'B = m_e \frac{v'^2}{R} \quad (v' > v)$$

$$ev'B = \frac{m_e}{R} (v'^2 - v^2) = \frac{m_e}{R} (v' + v)(v' - v)$$

assuming $\Delta v = v' - v$ is small,

$$\Delta v = \frac{eRB}{2m_e}$$

$$\Delta\vec{m} = -\frac{1}{2} e(\Delta v)R \hat{z} = -\frac{e^2 R^2}{4m_e} \vec{B}$$

P18. Boundary-Condition Reference Notes

(pp. 33–35)

[reference notes, not a worked problem]

Summaries of the boundary conditions (and asymptotic conditions) used to fix the coefficients in the classic separation-of-variables sphere problems — electrostatic potential V for charged/dielectric/conducting spheres, magnetic scalar potential W for magnetized and permeable spheres, and the ansatz for a conducting sphere with an insulating dielectric shell.

Solution.

Charge Density of Shell (conditions listed (ii) then (i) on the page):

$$(ii) \quad \left. \frac{\partial V_{in}}{\partial r} \right|_R - \left. \frac{\partial V_{out}}{\partial r} \right|_R = \frac{\sigma_0(\theta)}{\epsilon_0}$$

$$(i) \quad V_{out}|_R = V_{in}|_R$$

Dielectric Sphere in $\vec{E}$ -field:

$$(i) \quad V_{in}|_R = V_{out}|_R$$

$$(ii) \quad \epsilon \left. \frac{\partial V_{in}}{\partial r} \right|_{r=R} = \epsilon_0 \left. \frac{\partial V_{out}}{\partial r} \right|_R$$

$$(iii) \quad V_{out} \longrightarrow -E_0 r \cos \theta \quad (r \gg R)$$

Uncharged Sphere in E-field:

$$(i) \quad V_{in}|_R = V_{out}|_R$$

$$(ii) \quad -\epsilon_0 \left. \frac{\partial V}{\partial r} \right|_R = \sigma(\theta)$$

$$(iii) \quad V_{out} \longrightarrow -E_0 r \cos \theta \quad (r \gg R)$$

Uniform $\vec{M}$ sphere:

$$(i) \quad W_{in}|_R = W_{out}|_R$$

$$(ii) \quad \left. \frac{\partial W_{in}}{\partial r} \right|_R - \left. \frac{\partial W_{out}}{\partial r} \right|_R = M^\perp = M \hat{z} \cdot \hat{r} = M \cos \theta$$

Uniform $\vec{M}$ in B-field (the conditions are those of a linear permeable sphere in a uniform B_0):

$$(i) \quad W_{in}|_R = W_{out}|_R$$

$$(ii) \quad \mu \frac{\partial W_{in}}{\partial r} \Big|_R - \mu_0 \frac{\partial W_{out}}{\partial r} \Big|_R = 0$$

$$(iii) \quad W_{out} \longrightarrow -\frac{1}{\mu_0} B_0 r \cos \theta \quad (r \gg R)$$

Uncharged Cond. Sphere w/ insulating shell in E_0 -field:

$$V_{out} = -E_0 r \cos \theta + \sum_{\ell} \frac{B_{\ell}}{r^{\ell+1}} P_{\ell}(\cos \theta)$$

$$V_{int} = \sum_{\ell} \left(A_{\ell} r^{\ell} + \frac{\bar{B}_{\ell}}{r^{\ell+1}} \right) P_{\ell}(\cos \theta)$$

$$V_{in} = 0$$

$$V_{out} = V_{int} \quad (r = b)$$

$$\varepsilon \frac{\partial V_{int}}{\partial r} \Big|_b = \epsilon_0 \frac{\partial V_{out}}{\partial r} \Big|_b \quad V_{int}|_a = 0$$

P19. Ex 2.7 Potential of Spherical Shell w/ Uniform Surface Charge

(p. 36)

Potential inside and outside a spherical shell of radius R carrying total charge q uniformly on its surface, found by integrating $-\int \vec{E} \cdot d\vec{l}$ in from infinity.

Solution.

If the shell has a total charge q it can be treated as a point on the origin (from $r \geq R$).

$$\vec{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

[Diagram: sphere of radius R with a field point P outside on a radial ray.]

Due to symmetry

$$\vec{E}_{in} = 0$$

$$V_{out} = - \int_{\infty}^r \vec{E} \cdot d\vec{l} = - \frac{1}{4\pi\epsilon_0} q \int_{\infty}^r \frac{1}{r'^2} dr'$$

$$= + \frac{q}{4\pi\epsilon_0} \frac{1}{r'} \Big|_{\infty}^r = \frac{q}{4\pi\epsilon_0} \frac{1}{r}$$

$$V_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

(boxed) [Small sketch: the $1/r$ fall-off of V outside the shell, axis labeled r , marked " $R \rightarrow \infty$ ".]

$$V_{in}(r) = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^R \frac{q}{r'^2} dr' - \int_R^r 0 = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^R$$

$$\nabla V = 0$$

$$V_{in} = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

(boxed)

P20. Ex 4.8 Point Charge Above Linear Dielectric

(pp. 37-39)

Given: the region $z < 0$ is filled with a uniform linear dielectric of susceptibility χ_e .
Solve: calculate the force on a point charge q situated a distance d above the origin — via the bound surface charge σ_b at $z = 0$ and the method of images.

Solution.

[Diagram: 3-D sketch of the dielectric half-space $z < 0$ with axes x, y, z ; the charge q sits on the $\hat{z}$ -axis a height d above the origin, with angle θ to a surface point; annotation with an arrow to the interface: " $q(\text{surface}) = -q$ @ $z = 0$, so attractive".]

$$\sigma_b = \vec{P} \cdot \hat{n} = P_z = \epsilon_0 \chi_e E_z$$

[Diagram: right triangle with vertical leg d , horizontal leg r , hypotenuse $\sqrt{r^2 + d^2} = r$, angle θ at the top] — so when $d = r \cos \theta$,

$$-\frac{1}{4\pi\epsilon_0} \frac{q \cos \theta}{(r^2 + d^2)} = -\frac{1}{4\pi\epsilon_0} \frac{qd}{(r^2 + d^2)^{3/2}}$$

z -component of the field of the bound charge is $-\sigma_b/2\epsilon_0$ (pg 90 footnote).

$$\sigma_b = \epsilon_0 \chi_e \left(-\frac{1}{4\pi\epsilon_0} \frac{qd}{(r^2 + d^2)^{3/2}} - \frac{\sigma_b}{2\epsilon_0} \right)$$

then

$$\sigma_b = -\frac{1}{2\pi} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{qd}{(r^2 + d^2)^{3/2}}$$

ignoring $\left(\frac{\chi_e}{\chi_e + 2} \right)$ this is the same as the induced charge on an infinite conducting plane.

Total bound charge

$$q_b = - \left(\frac{\chi_e}{\chi_e + 2} \right) q$$

(boxed). By direct integration

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \sigma_b da$$

Method of images: replace the dielectric by a single point charge q_b at image position $(0, 0, -d)$:

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{q_b}{\sqrt{x^2 + y^2 + (z+d)^2}} \right) \quad \text{for } z > 0$$

A charge $q + q_b$ at $(0, 0, d)$ yields the potential *[preceding words struck out; this is the potential in the region $z < 0$]*

$$V = \frac{1}{4\pi\epsilon_0} \frac{q + q_b}{\sqrt{x^2 + y^2 + (z-d)^2}}$$

Using the two potentials, constitute a function that satisfies Poisson's equation with a point charge q at $(0, 0, d)$, which goes to zero at infinity, which is continuous at the boundary $z = 0$, and whose normal derivative exhibits the discontinuity appropriate to the surface charge σ_b at $z = 0$:

$$-\epsilon_0 \left(\left. \frac{\partial V}{\partial z} \right|_{z=0^+} - \left. \frac{\partial V}{\partial z} \right|_{z=0^-} \right) = -\frac{1}{2\pi} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{qd}{(x^2 + y^2 + d^2)^{3/2}}$$

The force on q is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_b}{(2d)^2} \hat{z} = -\frac{1}{4\pi\epsilon_0} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{q^2}{4d^2} \hat{z}$$

[Top of p. 39 carries a struck-through false start of the next problem's title ("Uniform Current Over a Wire"), taken up on p. 40.]

P21. Ex 5.4(a) Uniform Current Over a Wire

(pp. 40–41)

Griffiths Ex 5.4: (a) a current I is uniformly distributed over a wire of circular cross section with radius a — find the volume current density J ; (b) the current density in the wire is proportional to the distance from the axis, $J = ks$ — find the total current. The entry closes by deriving the continuity equation from the divergence theorem.

Solution.

Find: volume current density J .

$$J = \frac{I}{\pi a^2}$$

b) Current density in the wire is proportional:

$$J = ks$$

Solution:

$$\vec{J} = \frac{d\vec{I}}{da_{\perp}}, \quad da_{\perp} = s ds d\phi$$

$$I = \int J da_{\perp} = \iiint (ks)(s ds d\phi)$$

$$I = 2\pi k \int_0^a s^2 ds = \frac{2\pi k a^3}{3}$$

$$I = \int_S J da_{\perp} = \int_S \vec{J} \cdot d\vec{a}$$

Then by the divergence theorem,

$$\oint_S \vec{J} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{J}) d\tau$$

$$\int_V (\nabla \cdot \vec{J}) d\tau = -\frac{d}{dt} \int_V \rho d\tau = -\int_V \left(\frac{\partial \rho}{\partial t} \right) d\tau$$

giving the boxed continuity equation

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$q \sim \lambda dl \sim \sigma da \sim \rho d\tau$$

P22. Prob 4.12 Potential of Uniform Polarized Sphere

(p. 42)

Griffiths Prob 4.12: compute the potential of a uniformly polarized sphere (radius R , uniform $\vec{P}$) directly from $V = \frac{1}{4\pi\epsilon_0} \int \vec{P} \cdot \hat{\mathbf{r}}/r^2 d\tau'$, by recognizing the integral as the field of a uniformly charged sphere.

Solution.

$$\begin{aligned}
 V &= \frac{1}{4\pi\epsilon_0} \int \frac{\vec{P} \cdot \hat{\mathbf{r}}}{r^2} d\tau' \\
 &\rightarrow \frac{\vec{P}}{4\pi\epsilon_0} \cdot \int \frac{\vec{\mathbf{r}}}{r^3} d\tau, \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho \vec{\mathbf{r}}}{r^3} d\tau' \\
 V &= \vec{P} \cdot \frac{\vec{E}_{ucs}}{\rho}
 \end{aligned}$$

($\vec{E}_{ucs} = \vec{E}$ of the uniform charged sphere.)

$$\vec{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \quad \text{where } q = \rho \forall$$

$$\vec{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi R^3 \rho}{r^2} \hat{\mathbf{r}}$$

$$\vec{E}_{in} = \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi R^3 \rho}{R^3} \vec{\mathbf{r}}$$

Boxed results:

$$V_{in} = \vec{P} \cdot \frac{1}{3\epsilon_0} \vec{\mathbf{r}} = \frac{Pr \cos \theta}{3\epsilon_0}$$

$$V_{out} = \vec{P} \cdot \frac{R^3 \hat{\mathbf{r}}}{3\epsilon_0 r^2} = \frac{R^3 P \cos \theta}{3\epsilon_0 r^2}$$

P23. B Field of Perfect Dipole

(pp. 43–44)

Compute the magnetic field of a perfect (point) magnetic dipole m at the origin, directed along $\hat{z}$, from its vector potential by taking the curl in spherical coordinates.

[Diagram: x, y, z axes with dipole moment m along $\hat{z}$ at the origin; a field point at (r, θ, ϕ) .]

Solution.

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^2} \hat{\phi}$$

$$\vec{B}_{dip}(\vec{r}) = \nabla \times \vec{A}$$

curl in spherical

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta A_\phi) - \partial_\phi A_\theta] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \partial_\phi A_r - \partial_r (r A_\phi) \right] \hat{\theta} + \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\theta) - \partial_\theta A_r \right) \hat{\phi}$$

(the $\partial_\phi A_\theta$ and $\partial_\phi A_r$ terms are marked zero and the whole $\hat{\phi}$ term is struck out, since only A_ϕ is nonzero)

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} (\partial_\theta (\sin \theta A_\phi)) \hat{r} - \frac{1}{r} \partial_r (r A_\phi) \hat{\theta}$$

$$\partial_\theta (\sin \theta A_\phi) = \frac{\mu_0}{4\pi} \frac{m}{r^2} 2 \sin \theta \cos \theta$$

$$\partial_r (r A_\phi) = \frac{\mu_0}{4\pi} m \sin \theta \partial_r (r^{-1}) \rightarrow \frac{\mu_0}{4\pi} m \sin \theta \left(-\frac{1}{r^2} \right)$$

$$\nabla \times \vec{A} = \frac{\mu_0}{4\pi} \frac{m}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

Boxed:

$$\vec{B}_{dip} = \frac{\mu_0 m}{4\pi r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

P24. Ex 4.2 Uniformly Polarized Sphere

(p. 45)

Griffiths Ex 4.2: the field of a uniformly polarized sphere of radius R , obtained from the bound surface charge $\sigma_b = P \cos \theta$ by reusing the $\sigma = k \cos \theta$ boundary-value result; inside the field is $-\vec{P}/3\epsilon_0$, outside it is a perfect dipole.

Solution.

$$\sigma_b = \vec{P} \cdot \hat{n} = P \cos \theta$$

$$V_{in} = \frac{k}{3\epsilon_0} r \cos \theta \implies \frac{Pr \cos \theta}{3\epsilon_0}$$

$$V_{out} = \frac{kR^3}{3\epsilon_0 r^2} \longrightarrow \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta$$

$$z = r \cos \theta$$

$$\vec{E}_{in} = -\nabla V = -\nabla \frac{Pz}{3\epsilon_0} \hat{z} = -\frac{\vec{P}}{3\epsilon_0}$$

Outside can be treated as perfect dipole

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2} \quad r \geq R$$

$$\vec{p} = \vec{P} d\tau'$$

which agrees since $\vec{p} = \frac{4}{3}\pi R^3 \vec{P}$

$$V = \frac{4\pi R^3}{3 \cdot 4\pi\epsilon_0} \frac{\vec{P} \cdot \hat{r}}{r^2} = \frac{PR^3 \cos \theta}{3\epsilon_0 r^2}$$

P25. Ex 2.3 Uniform Charged Sphere

(p. 46)

Solve: $\vec{E}$ outside a uniformly charged sphere of charge q with radius R , by Gauss's law; then rewritten in terms of ρ , together with the interior field.

[Diagram: sphere of radius R inside a larger concentric dashed spherical Gaussian surface of radius r , labeled "Pill Box".]

Solution.

Gaussian surface @ ($r > R$):

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}, \quad Q_{enc} = q$$

$$\int_S \vec{E} \cdot d\vec{a} = \int |\vec{E}| da = |\vec{E}| 4\pi r^2$$

Boxed:

$$\vec{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$\rho = \frac{q}{V}$$

$$E_{out} = \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi R^3 \rho}{r^2} \hat{r}$$

$$\vec{E}_{out} = \frac{\rho R^3}{3\epsilon_0 r^2} \hat{r}, \quad \vec{E}_{in} = \frac{\rho}{3\epsilon_0} \vec{r}$$

P26. Magnetostatics

(p. 47)

[reference notes, not a worked problem]

Opening reference sheet for the magnetostatics section: the magnetic and electric forces on a charge, the Lorentz force law, the statement that magnetic forces do no work, and the line-current form of the magnetic force.

[Diagram: B field circling a wire carrying current I upward; two antiparallel currents I with the field lines between them bulging apart; two parallel currents I bowing toward each other with force F .]

Solution.

$$\vec{F}_{mag} = Q \left(\vec{v} \times \vec{B} \right), \quad \vec{F}_E = Q\vec{E}$$

Lorentz Force Law

$$\vec{F} = Q \left[\vec{E} + \left(\vec{v} \times \vec{B} \right) \right]$$

magnetic forces do no work

$$dW_{mag} = \vec{F}_{mag} \cdot d\vec{l} = Q \left(\vec{v} \times \vec{B} \right) \cdot \vec{v} dt = 0$$

Currents:

$$\vec{I} = \lambda \vec{v}$$

$$\vec{F}_{mag} = \int \left(\vec{v} \times \vec{B} \right) dq = \int \left(\vec{v} \times \vec{B} \right) \lambda dl = \int \left(\vec{I} \times \vec{B} \right) dl$$

$$\vec{F}_{mag} = \int I \left(d\vec{l} \times \vec{B} \right)$$

P27. Ex 2.4 Cylindrically Charged Density

(p. 48)

Griffiths Ex 2.4: a long cylinder carries charge density proportional to the distance from the axis, $\rho = ks$; find $\vec{E}$ inside using a Gaussian cylinder of radius s and length ℓ .

[Diagram: cylinder of length ℓ about the $\hat{z}$ axis, with s the radial distance from the axis.]

Solution.

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}, \quad A_{cylinder} = 2\pi s\ell, \quad d\tau = s' ds' d\phi dz$$

$$Q_{enc} = \int \rho d\tau = \int_0^{2\pi} d\phi \int_0^\ell dz \int_0^s ks'^2 ds'$$

$$Q_{enc} = 2\pi k\ell \int_0^s s'^2 ds' = 2\pi k\ell \frac{s^3}{3}$$

$$\oint_S |\vec{E}| da = \frac{Q_{enc}}{\epsilon_0}$$

$$|\vec{E}| 2\pi s\ell = \frac{1}{\epsilon_0} \frac{2}{3} \pi k\ell s^3$$

Boxed:

$$\vec{E}_{in} = \frac{ks^2}{3\epsilon_0} \hat{s}$$

P28. Ex 2.5 Rectangular Pill Box

(p. 49)

Griffiths Ex 2.5: an infinite plane carries uniform surface charge σ ; find $\vec{E}$ using a rectangular Gaussian pillbox of face area A straddling the plane.

[Diagram: infinite plane with a rectangular pillbox of face area A straddling it; $\vec{E}$ points up above the plane and down below.]

Solution.

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}, \quad Q_{enc} = \sigma A$$

$$\oint \vec{E} \cdot d\vec{a} = \int |\vec{E}| da = 2A |\vec{E}|$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow 2A |\vec{E}| = \frac{\sigma A}{\epsilon_0}$$

Boxed:

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$$

P29. Ex 4.4 Wire with Rubber Dielectric

(p. 50)

Griffiths Ex 4.4: a uniform line charge λ with rubber insulation up to $r = a$; find $\vec{D}$ from the free charge using a Gaussian cylinder of radius s and length L , then $\vec{E}$ outside the rubber.

[Diagram: wire carrying line charge λ along the axis of a dielectric cylinder of radius a , with a coaxial Gaussian cylinder of radius s and length L inside it.]

Solution.

$$\oint \vec{D} \cdot d\vec{a} = Q_{f\,enc}, \quad \vec{D} \equiv \epsilon_0 \vec{E} + \vec{P}$$

$$A_{cyl} = 2\pi r h, \quad A_{gaus} = 2\pi s L$$

$$\oint \vec{D} \cdot d\vec{a} \longrightarrow D(2\pi s L), \quad Q_{f\,enc} \longrightarrow \lambda L$$

$$D 2\pi s L = \lambda L$$

Boxed:

$$\vec{D} = \frac{\lambda}{2\pi s} \hat{s}$$

for $s > a$, $\vec{P} = 0$, and $\vec{D} \equiv \epsilon_0 \vec{E} + \vec{P}$ with the $\vec{P}$ term marked zero, so boxed:

$$\vec{E}_{out} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s} \quad (s > a)$$

P30. Ex 7.2

(p. 51)

Griffiths Ex 7.2: the radial current between two coaxial cylinders (inner radius a , outer radius b , length L) separated by material of conductivity σ , starting from the line-charge field $\vec{E} = \lambda/(2\pi\epsilon_0 s) \hat{s}$; relate I to the potential difference V .

[Diagram: coaxial cylinders of inner radius a and outer radius b , length L .]

Solution.

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$$

$$I = \int \vec{J} \cdot d\vec{a} = \sigma \int \vec{E} \cdot d\vec{a} = \frac{\lambda\sigma}{2\pi\epsilon_0 s} (2\pi s L)$$

$$I = \frac{\lambda\sigma L}{\epsilon_0}$$

$$V = - \int_b^a \vec{E} \cdot d\vec{l} = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

$$I = \frac{2\pi\sigma L}{\ln\left(\frac{b}{a}\right)} V$$

P31. Uniform Line Charge / Solenoid / Magnetized Cylinder and Sphere

(pp. 52-54)

[reference notes, not a worked problem]

Quick-reference results: Gauss's law for a uniform line charge and inside a uniformly charged sphere; the on-axis solenoid integral with the $z = a \cot \theta$ substitution; nested and single solenoids; and the uniformly magnetized cylinder and sphere.

Solution.

Uniform line charge. *[Diagram: short cylinder of length ℓ enclosing a line charge λ .]*

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

$$\vec{E} \cdot 2\pi s L = \frac{\lambda L}{\epsilon_0}$$

$$\vec{E} = \frac{\lambda}{2\pi s \epsilon_0} \hat{s}$$

Unif. Charged [sphere]. *[Diagram: sphere.]*

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0} = \frac{1}{\epsilon_0} \int \rho d\tau$$

$$\vec{E} \cdot 4\pi r^2 = \frac{\frac{4}{3}\pi R^3 \rho}{\epsilon_0}$$

[sic: for the interior field the enclosed charge should be $\frac{4}{3}\pi r^3 \rho$; the boxed result below corresponds to r^3]

$$\vec{E} = \frac{1}{3\epsilon_0} \rho r \hat{r}$$

On-axis solenoid field. *[Diagram: cylinder of radius a with a circumferential current loop; right triangle with legs a and z , hypotenuse r , giving $z = a \cot \theta$.]*

$$B = \frac{\mu_0 n I}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz, \quad dz = -\frac{a}{\sin^2 \theta} d\theta$$

Nested solenoids. *[Diagram: two coaxial solenoid windings drawn with oppositely directed winding arrows.]*

$$B_{mid} = \mu_0 I n_2 \hat{z}, \quad B_{out} = 0$$

$$B_{in} = \mu_0 I (n_2 - n_1) \hat{z}$$

Solenoid:

$$B_{in} = \mu_0 n I \hat{z}, \quad B_{out} = 0$$

Cylinder. *[Diagram: cylinder with $\vec{M}$ along its axis.]*

$$\vec{J}_b = \nabla \times \vec{M} = 0, \quad \vec{K}_b = \vec{M} \times \hat{n} = M \hat{\phi}$$

$$B_{out} = 0, \quad B = \mu_0 K_b = \mu_0 M$$

$$\vec{B} = \mu_0 \vec{M}$$

Magnetized Cylinder: Long $\rightarrow$ Solenoid; Short $\rightarrow$ Dipole.

Magnetized Sphere:

$$\vec{B} = \frac{2}{3} \mu_0 \vec{M}, \quad \vec{m} = \frac{4}{3} \pi R^3 \vec{M}$$

P32. Energy in Magnetic Fields

(pp. 55-56)

Derivation of the energy stored in an inductor and in the magnetic field: from $dW/dt = -\mathcal{E}I$ to $W = \frac{1}{2}LI^2$, then rewriting via the vector potential and product rule 6 to $W = \frac{1}{2\mu_0} \int B^2 d\tau$; the back of the page collects the electric and magnetic energy formulas side by side.

Solution.

$$\frac{dW}{dt} = -\mathcal{E}I = LI \frac{dI}{dt}$$

Boxed:

$$W = \frac{1}{2}LI^2$$

$$\Phi = \int \vec{B} \cdot d\vec{a} = \int (\nabla \times \vec{A}) \cdot d\vec{a} = \oint \vec{A} \cdot d\vec{l}$$

$$LI = \oint \vec{A} \cdot d\vec{l}$$

$$W = \frac{1}{2}I \oint \vec{A} \cdot d\vec{l} = \frac{1}{2} \oint (\vec{A} \cdot \vec{I}) dl$$

$$W = \frac{1}{2} \int_{\forall} (\vec{A} \cdot \vec{J}) d\tau \quad (\nabla \times \vec{B} = \mu_0 \vec{J})$$

$$W = \frac{1}{2\mu_0} \int \vec{A} \cdot (\nabla \times \vec{B}) d\tau$$

product rule 6

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

$$\vec{A} \cdot (\nabla \times \vec{B}) \rightarrow \vec{B} \cdot \vec{B} - \nabla \cdot (\vec{A} \times \vec{B})$$

$$W = \frac{1}{2\mu_0} \int_{All} B^2 d\tau$$

more on back:

$$W_{elec} = \frac{1}{2} \int (V\rho) d\tau = \frac{\epsilon_0}{2} \int E^2 d\tau$$

$$W_{mag} = \frac{1}{2} \int (\vec{A} \cdot \vec{J}) d\tau = \frac{1}{2\mu_0} \int B^2 d\tau$$

P33. Ex 3.8 Uncharged Metal Sphere in Uniform E-field

(pp. 57-59)

Griffiths Ex 3.8: an uncharged metal sphere of radius R is placed in a uniform field $\vec{E} = E_0 \hat{z}$; find the potential outside by separation of variables in Legendre polynomials, then the induced surface charge density.

[Diagram: sphere of radius R ; uniform field lines along $\hat{z}$ bending near the sphere, with induced + charge drawn on the top surface and - on the bottom, regions labeled E_- and E_+ ; axes $\hat{z}$, $\hat{y}$.]

Solution.

By symmetry x-y plane potential is zero, $V = 0$.

at large z

$$V \rightarrow -E_0 z + C$$

from xy-plane $V = 0$, so $C = 0$.

at $z = 0$, with $z = r \cos \theta$:

$$\textcircled{1} V = 0 \quad (r = R)$$

$$\textcircled{2} V \rightarrow -E_0 r \cos \theta \quad (r \gg R)$$

1st case:

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_{\ell} r^{\ell} + \frac{B_{\ell}}{r^{\ell+1}} \right) P_{\ell}(\cos \theta)$$

$\textcircled{1}$ at $z = 0$ focuses on the radial factor [*underbrace beneath* $(A_{\ell} r^{\ell} + B_{\ell}/r^{\ell+1})$] so that $V = 0$ at $r = R$:

$$A_{\ell} R^{\ell} + \frac{B_{\ell}}{R^{\ell+1}} = 0$$

$$B_{\ell} = -A_{\ell} R^{2\ell+1}$$

$$V(r, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} \left(r^{\ell} - \frac{R^{2\ell+1}}{r^{\ell+1}} \right) P_{\ell}(\cos \theta)$$

Condition $\textcircled{2}$ for $r \gg R$: second term disappears, so that where ($V \approx -E_0 r \cos \theta$)

$$V(r, \theta) \rightarrow \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta) = -E_0 r \cos \theta$$

$$A_0 r^0 P_0 + A_1 r P_1(\cos \theta) + A_2 r^2 P_2(\cos \theta) = -E_0 r \cos \theta$$

Matching term by term (ℓ | LHS = RHS): for $\ell = 0$, $A_0 = 0$; for $\ell = 1$, $A_1 r \cos \theta = -E_0 r \cos \theta$; for $\ell = 2$, $A_2 r^2 P_2(\cos \theta) = 0$.

$$A_1 r \cos \theta = -E_0 r \cos \theta$$

$$A_1 = -E_0 \quad \text{so} \quad \ell = 1$$

Boxed:

$$V(r, \theta) = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos \theta$$

first-component: external field; 2nd-component: induced charge.

Induced Charge density

$$\sigma(\theta) = -\epsilon_0 \left. \frac{\partial V}{\partial r} \right|_{r=R}$$

$$-\frac{\partial V}{\partial r} = +E_0 \left(1 + \frac{2R^3}{r^3} \right) \cos \theta$$

$$\sigma(\theta) = -\epsilon_0 \left. \frac{\partial V}{\partial r} \right|_{r=R} = 3E_0 \epsilon_0 \cos \theta$$

Boxed:

$$\sigma(\theta) = 3E_0 \epsilon_0 \cos \theta$$

P34. Ex 3.9 Charge Density on Spherical Shell

(pp. 60-61)

Find the potential inside and outside a spherical shell of radius R carrying surface charge $\sigma_0(\theta) = k \cos \theta$, by separation of variables and the two boundary conditions at $r = R$. *[The header first reads "Ex 4.7 Homogeneous Linear Dielectric sphere in E-field", struck through and overwritten with this title; that dielectric problem is worked on pp. 62-65.]*

Solution.

General solutions inside and outside:

$$V_{in} = \sum A_\ell r^\ell P_\ell(\cos \theta)$$

$$V_{out} = \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta)$$

[Margin notes: $\nabla \cdot \vec{E} = \rho/\epsilon_0$; $E_\parallel$ continuous; $\vec{E} = -\nabla V$.]

(ii) The derivative jump at the shell:

$$\left. \frac{\partial V_{out}}{\partial r} \right|_R - \left. \frac{\partial V_{in}}{\partial r} \right|_R = -\frac{1}{\epsilon_0} \sigma_0(\theta)$$

$$\sum -\frac{(\ell+1)B_\ell}{R^{\ell+2}} P_\ell(\cos \theta) - \sum \ell A_\ell R^{\ell-1} P_\ell(\cos \theta) = -\frac{\sigma_0(\theta)}{\epsilon_0}$$

(i) Continuity of the potential:

$$V_{out}|_R = V_{in}|_R$$

$$\sum A_\ell R^\ell P_\ell(\cos \theta) = \sum \frac{B_\ell}{R^{\ell+1}} P_\ell(\cos \theta)$$

Matching coefficients (the notes tag this line $\ell \neq 1$) *[sic: this match holds for every ℓ , including $\ell = 1$]:*

$$A_\ell R^\ell = \frac{B_\ell}{R^{\ell+1}} \rightarrow B_\ell = A_\ell R^{2\ell+1}$$

Substituting into (ii):

$$\sum -(\ell+1)A_\ell \frac{R^{2\ell+1}}{R^{\ell+2}} P_\ell(\cos \theta) - \ell A_\ell R^{\ell-1} P_\ell(\cos \theta) = -\frac{\sigma_0(\theta)}{\epsilon_0}$$

$$\sum_{\ell=0}^{\infty} (2\ell + 1) A_{\ell} R^{\ell-1} P_{\ell}(\cos \theta) = \frac{\sigma_0(\theta)}{\epsilon_0}$$

since $\sigma_0(\theta) \equiv k \cos \theta$:

($\ell = 1$)

$$3A_1 \cos \theta = \frac{k \cos \theta}{\epsilon_0} \rightarrow A_1 = \frac{k}{3\epsilon_0}$$

($\ell \neq 1$) $\rightarrow A_{\ell} \rightarrow$ zero. With $B_1 = A_1 R^3$, the boxed results:

$V_{in} = \frac{k}{3\epsilon_0} r \cos \theta \qquad V_{out} = \frac{k}{3\epsilon_0} \frac{R^3 \cos \theta}{r^2}$

P35. Ex 4.7 Homogeneous Linear Dielectric Sphere in E_0 -Field

(pp. 62-65)

A homogeneous linear dielectric sphere of radius R (permittivity ϵ , $\epsilon_r = \epsilon/\epsilon_0$) sits in an otherwise uniform field $\vec{E}_0$; find the potential everywhere and the field inside, by separation of variables with the dielectric boundary conditions.

Solution.

[Diagram: vertical field lines labeled E_0 threading and bending around a sphere; short arrows inside the sphere indicate the uniform interior field.]

Boundary conditions:

$$(i) V_{in}|_{r=R} = V_{out}|_{r=R}, (ii) \epsilon \frac{\partial V_{in}}{\partial r} \Big|_{r=R} = \epsilon_0 \frac{\partial V_{out}}{\partial r} \Big|_{r=R}, (iii) V_{out} \rightarrow -E_0 r \cos \theta \quad (r \gg R).$$

$$V_{in} = \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta)$$

$$V_{out} = -E_0 r \cos \theta + \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{r^{\ell+1}} P_{\ell}(\cos \theta)$$

(i):

$$\sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} P_{\ell}(\cos \theta) = -E_0 R \cos \theta + \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{R^{\ell+1}} P_{\ell}(\cos \theta)$$

let $\ell = 1$ (the $\cos \theta$'s cancel):

$$A_1 R = -E_0 R + \frac{B_1}{R^2}$$

($\ell \neq 1$):

$$A_{\ell} R^{\ell} = \frac{B_{\ell}}{R^{\ell+1}}$$

[Margin: $\epsilon_r = \epsilon/\epsilon_0$.]

(ii):

$$\frac{\partial V_{in}}{\partial r} \Big|_{r=R} = \sum_{\ell=0}^{\infty} A_{\ell} \ell r^{\ell-1} P_{\ell}(\cos \theta) \Big|_{r=R}$$

$$\frac{\partial V_{out}}{\partial r} \Big|_{r=R} = -E_0 \cos \theta - \sum_{\ell=0}^{\infty} (\ell+1) \frac{B_{\ell}}{R^{\ell+2}} P_{\ell}(\cos \theta)$$

$$\epsilon \left. \frac{\partial V_{in}}{\partial r} \right|_{r=R} = \epsilon_0 \left. \frac{\partial V_{out}}{\partial r} \right|_{r=R}$$

$$\epsilon_r \sum_{\ell=0}^{\infty} \ell A_{\ell} R^{\ell-1} P_{\ell}(\cos \theta) = -E_0 \cos \theta - \sum_{\ell=0}^{\infty} \frac{(\ell+1)B_{\ell}}{R^{\ell+2}} P_{\ell}(\cos \theta)$$

($\ell = 1$), cancelling $\cos \theta$:

$$\epsilon_r A_1 = -E_0 - \frac{2B_1}{R^3}$$

($\ell \neq 1$), cancelling P_{ℓ} :

$$\epsilon_r \ell A_{\ell} R^{\ell-1} = -(\ell+1) \frac{B_{\ell}}{R^{\ell+2}}$$

[Margin bracket: $\ell = 0 \rightarrow B_{\ell} = 0, A_0 = 0$; $\ell \rightarrow \infty$: $A_{\ell} \rightarrow 0, B_{\ell} \rightarrow 0$.]

where, from (i), $A_1 R = -E_0 R + \frac{B_1}{R^2}$, i.e. (cancelling one R and correcting the exponent)

$$(A_1 + E_0) = \frac{B_1}{R^3}$$

$$\epsilon_r A_1 = -E_0 - 2(A_1 + E_0)$$

$$\epsilon_r A_1 + 2A_1 = -3E_0$$

$$A_1 (\epsilon_r + 2) = -3E_0$$

$$\boxed{A_1 = \frac{-3E_0}{2 + \epsilon_r}}$$

Returning to $\epsilon_r A_1 = -E_0 - 2B_1/R^3$ with this A_1 [a false start beside it is scribbled out]:

$$\frac{-3E_0}{2 + \epsilon_r} \epsilon_r = -E_0 - \frac{2B_1}{R^3}$$

$$\frac{3E_0 \epsilon_r}{2 + \epsilon_r} - E_0 = \frac{2B_1}{R^3}$$

$$E_0 \left(\frac{3\epsilon_r}{2 + \epsilon_r} - 1 \right) \frac{R^3}{2} = B_1$$

$$E_0 \left(\frac{3\epsilon_r - (2 + \epsilon_r)}{2 + \epsilon_r} \right) \frac{R^3}{2} = B_1$$

$$\frac{E_0 R^3}{2} \left(\frac{2\epsilon_r - 2}{\epsilon_r + 2} \right) = B_1$$

cancelling the 2's:

$$\boxed{B_1 = E_0 R^3 \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right)}$$

$$\boxed{V_{in} = \frac{-3E_0}{2 + \epsilon_r} r \cos \theta = -\frac{3E_0}{2 + \epsilon_r} z}$$

$$\vec{E} = -\vec{\nabla} V:$$

$$\boxed{\vec{E}_{in} = \frac{3E_0}{2 + \epsilon_r}}$$

P36. Prob 4.23

(p. 66)

Find the field inside a sphere of linear dielectric (susceptibility χ_e) in an otherwise uniform field $\vec{E}_0$ by successive approximations: each polarization produces a uniform field $-\vec{P}/3\epsilon_0$ inside the sphere, which repolarizes the material, and the series is summed. *[Title line mostly hidden under a folded page corner; only "4.23" is clearly legible.]*

Solution.

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{E}_1 = -\frac{1}{3\epsilon_0} \vec{P}_0 = -\frac{\chi_e}{3} \vec{E}_0$$

$$\vec{P}_1 = \epsilon_0 \chi_e \vec{E}_1 = -\epsilon_0 \frac{\chi_e^2}{3} \vec{E}_0$$

$$\vec{E}_2 = -\frac{1}{3\epsilon_0} \vec{P}_1 = \frac{\chi_e^2}{9} \vec{E}_0$$

$$\vec{E} = \vec{E}_0 + \vec{E}_1 + \vec{E}_2 + \cdots = \sum_{n=0}^{\infty} \left(-\frac{\chi_e}{3}\right)^n \vec{E}_0$$

where

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$\vec{E} = \frac{\vec{E}_0}{\left(1 + \frac{\chi_e}{3}\right)}$$

P37. Ex 6.1 Uniformly Magnetized Sphere

(p. 67)

Find $\vec{B}$ of a uniformly magnetized sphere ($\vec{M}$ along $\hat{z}$, radius R) from its bound currents, using the equivalence to a spinning charged spherical shell.

Solution.

[Diagram: sphere with a $\hat{z}$ axis and magnetization $\vec{M}$ inside.]

$$\vec{J}_b = \nabla \times \vec{M} = 0$$

$$\vec{K}_b = \vec{M} \times \hat{n} = M \sin \theta \hat{\phi}$$

identical to spinning spherical shell

$$\vec{K} = \sigma \vec{v} = \sigma \omega R \sin \theta \hat{\phi}, \quad \sigma R \vec{\omega} \equiv \vec{M}$$

where

$$\vec{B} = \frac{2}{3} \mu_0 \sigma R \vec{\omega}$$

so then

$$\boxed{\vec{B}_{in} = \frac{2}{3} \mu_0 \vec{M}} \quad \boxed{\vec{B}_{out} = \vec{B}_{dipole}}$$

$$\vec{m} = V \vec{M} = \boxed{\frac{4}{3} \pi R^3 \vec{M} = \vec{m}}$$

P38. 6.15 Uniformly Magnetized Sphere

(pp. 68–69)

The uniformly magnetized sphere ($\vec{M} = M\hat{z}$) again, now by separation of variables for the magnetic scalar potential W : since there is no free current, $\vec{H} = -\nabla W$, with the jump in $\partial W/\partial r$ set by $M_\perp$.

Solution.

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

(the $\partial \vec{D}/\partial t$ term is crossed out — zero here), and

$$\vec{B} = -\mu_0 \nabla \phi_m, \quad \phi_m \equiv W$$

[Margin: $\vec{B} = \mu_0(\vec{H} + \vec{M})$; $\vec{B} = \mu\vec{H}$.]

$$W_{in} = \sum A_\ell r^\ell P_\ell(\cos \theta), \quad W_{out} = \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta)$$

(i)

$$W_{in}|_R = W_{out}|_R$$

(ii)

$$-\left. \frac{\partial W_{out}}{\partial r} \right|_R + \left. \frac{\partial W_{in}}{\partial r} \right|_R = M_\perp = M \hat{z} \cdot \hat{r} = M \cos \theta$$

(i), cancelling $P_\ell(\cos \theta)$ on both sides:

$$A_\ell R^\ell = \frac{B_\ell}{R^{\ell+1}} \rightarrow B_\ell = A_\ell R^{2\ell+1}$$

(ii):

$$\frac{B_\ell(\ell+1)}{R^{\ell+2}} P_\ell(\cos \theta) + A_\ell \ell R^{\ell-1} P_\ell(\cos \theta) = M \cos \theta$$

$$A_\ell \frac{R^{2\ell+1}}{R^{\ell+2}} (\ell+1) P_\ell(\cos \theta) + A_\ell \ell R^{\ell-1} P_\ell(\cos \theta) = M \cos \theta$$

$\ell \neq 1$:

$$A_\ell \left[R^{\ell-1} (2\ell+1) P_\ell(\cos \theta) \right] = 0 \rightarrow A_\ell = 0$$

$\ell = 1$ (the $\cos \theta$'s cancel, $R^0 = 1$):

$$A_1 R^0(2) \cos \theta + A_1 R^0 \cos \theta = M \cos \theta$$

$$2A_1 + A_1 = M \quad A_1 = \frac{M}{3}$$

$$W_{in} = \frac{M}{3} r \cos \theta = \frac{M}{3} z$$

$$\vec{H}_{in} = -\nabla W_{in} = -\frac{M}{3} \hat{z}$$

[Margin: $\vec{B} \approx \mu_0 \vec{H}$; $\vec{B} = -\mu_0 \nabla \phi_m$; $\mu_0 \vec{H} = -\mu_0 \nabla W$; $\vec{H}_{in} \approx -\nabla W_{in}$.]

$$\vec{B} = \mu_0 (\vec{H} + \vec{M}) = \mu_0 \left(-\frac{1}{3}M + M \right) \hat{z} = \frac{2}{3} \mu_0 M \hat{z}$$

$$\vec{B}_{in} = \frac{2}{3} \mu_0 M \hat{z}$$

P39. 6.18 Uniformly Magnetized Sphere in B_0

(pp. 70-74)

A sphere of linear magnetic material (permeability μ , susceptibility χ_m ; the header calls it "uniformly magnetized") in an otherwise uniform field $\vec{B}_0 = B_0 \hat{z}$: find the new field inside. Solved first as a boundary-value problem for the magnetic scalar potential W , then re-derived by successive approximation of the magnetization.

Solution.

$$\vec{B} = -\mu_0 \nabla \phi_m, \quad \phi_m \equiv W, \quad \vec{H}_{in} \approx -\nabla W_{in}$$

$$V_{in} = \sum A_\ell r^\ell P_\ell(\cos \theta) \quad V_{out} = \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta)$$

b.c.s:

(i)

$$W_{in}|_R = W_{out}|_R$$

(ii)

$$\mu \frac{\partial W_{in}}{\partial r} \Big|_R = \mu_0 \frac{\partial W_{out}}{\partial r} \Big|_R$$

(iii) for $r \gg R$:

$$\vec{B} \rightarrow B_0 \hat{z}, \quad \vec{H} \approx \frac{1}{\mu_0} \vec{B} = \frac{B_0}{\mu_0} \hat{z}$$

[Margin: $H \approx -\nabla W$, so $W \approx -\int \vec{B} dz$.]

$$W_{out} \rightarrow -\frac{1}{\mu_0} B_0 r \cos \theta$$

$$W_{out} = -\frac{1}{\mu_0} B_0 r \cos \theta + \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos \theta)$$

(i):

$$\sum A_\ell R^\ell P_\ell(\cos \theta) = -\frac{1}{\mu_0} B_0 R \cos \theta + \sum \frac{B_\ell}{R^{\ell+1}} P_\ell(\cos \theta)$$

($\ell \neq 1$), cancelling P_ℓ :

$$A_\ell R^\ell = \frac{B_\ell}{R^{\ell+1}}$$

$$A_\ell R^{2\ell+1} = B_\ell \quad (1)$$

($\ell = 1$), cancelling $\cos \theta$, then multiplying through by R^2 :

$$A_1 R = -\frac{1}{\mu_0} B_0 R + \frac{B_1}{R^2}$$

$$A_1 R^3 = -\frac{1}{\mu_0} B_0 R^3 + B_1 \quad (2)$$

(ii):

$$\mu \left(A_\ell \ell R^{\ell-1} P_\ell(\cos \theta) \right) - \mu_0 \left[-\frac{1}{\mu_0} B_0 \cos \theta - \frac{(\ell+1)B_\ell}{R^{\ell+2}} P_\ell(\cos \theta) \right] = 0$$

($\ell \neq 1$):

$$\mu A_\ell \ell R^{\ell-1} P_\ell + \mu_0 \left[\frac{(\ell+1)B_\ell}{R^{\ell+2}} P_\ell \right] = 0$$

using (1):

$$\mu A_\ell \ell R^{\ell-1} + \mu_0 \frac{(\ell+1)}{R^{\ell+2}} A_\ell R^{2\ell+1} = 0$$

[a line is scribbled out]

$$A_\ell \left[\mu \ell R^{\ell-1} + \mu_0 (\ell+1) R^{\ell-1} \right] = 0$$

$$A_\ell = 0$$

(ii) ($\ell = 1$), cancelling $\cos \theta$:

$$\mu A_1 \cos \theta + \mu_0 \left[\frac{1}{\mu_0} B_0 \cos \theta + \frac{(1+1)B_1}{R^3} \cos \theta \right] = 0$$

$$\mu A_1 + B_0 + \mu_0 \frac{2B_1}{R^3} = 0 \quad (3)$$

From (2):

$$B_1 = A_1 R^3 + \frac{B_0}{\mu_0} R^3 \rightarrow B_1 = R^3 \left(A_1 + \frac{B_0}{\mu_0} \right)$$

(2) $\rightarrow$ (3):

$$\mu A_1 + B_0 + \frac{2\mu_0}{R^3} R^3 \left(A_1 + \frac{B_0}{\mu_0} \right) = 0$$

$$\mu A_1 + B_0 + 2\mu_0 A_1 + 2B_0 = 0$$

$$A_1 (\mu + 2\mu_0) + 3B_0 = 0$$

$$A_1 = \frac{-3}{\mu + 2\mu_0} B_0$$

$$W_{in} = \left(\frac{-3}{\mu + 2\mu_0} \right) B_0 r \cos \theta = \frac{-3}{\mu + 2\mu_0} B_0 z$$

$$\vec{H}_{in} \approx -\nabla W_{in} \rightarrow \frac{+3}{\mu + 2\mu_0} B_0 \hat{z}$$

$$\vec{B} = \mu \vec{H} \rightarrow \frac{3\mu}{\mu + 2\mu_0} B_0 \hat{z}$$

With $\mu_0 \mu_r = \mu$ and $\mu_r = 1 + \chi_m$ [*a scratched-out expression between*], i.e. $\mu = \mu_0(1 + \chi_m)$:

$$\vec{B} = \frac{3\mu_0 (1 + \chi_m)}{\mu_0 (1 + \chi_m) + 2\mu_0} B_0 \hat{z}$$

cancelling the μ_0 's:

$$\boxed{\vec{B} = \frac{1 + \chi_m}{\left(1 + \frac{\chi_m}{3}\right)} B_0 \hat{z}}$$

Second method (below a dividing line): successive magnetization.

$$\vec{M} = \chi_m \vec{H}, \quad \vec{M}_0 = \chi_m \vec{H}_0$$

$$[Margin: \frac{\vec{B}}{\mu_0 \mu_r} = \vec{H}; \vec{H} = \frac{\vec{B}}{\mu_0 (1 + \chi_m)}.]$$

$$\vec{M}_0 = \frac{\chi_m}{\mu_0 (1 + \chi_m)} \vec{B}_0, \quad K \equiv \frac{\chi_m}{1 + \chi_m}$$

$$\vec{B}_1 \equiv \vec{B}_{in} = \frac{2}{3}\mu_0\vec{M}_0 = \frac{2}{3}\frac{\chi_m}{1+\chi_m}\vec{B}_0 = \frac{2}{3}K\vec{B}_0$$

$\vec{B}_1 = \frac{2}{3}K\vec{B}_0$, which magnetizes an additional amt.

$$\vec{B}_2 = \frac{2}{3}\mu_0\vec{M}_1 = \frac{2}{3}K\vec{B}_1 = \left(\frac{2K}{3}\right)^2 \vec{B}_0$$

$$\vec{B} = \vec{B}_0 + \vec{B}_1 + \vec{B}_2 + \dots$$

$$\vec{B} = \vec{B}_0 + \left(\frac{2}{3}K\right)\vec{B}_0 + \left(\frac{2K}{3}\right)^2 \vec{B}_0$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{where } x \equiv \frac{2}{3}K$$

$$\vec{B} = \frac{\vec{B}_0}{1 - \frac{2}{3}K} = \frac{\vec{B}_0}{1 - \frac{2}{3}\left(\frac{\chi_m}{1+\chi_m}\right)}$$

$$\boxed{\vec{B} = \left(\frac{1+\chi_m}{1+\frac{\chi_m}{3}}\right)\vec{B}_0} \quad \left(\vec{B}_0 = B_0\hat{z}\right)$$

P40. Prob 4.5 Torque on Perfect Dipoles

(p. 75)

Two perfect dipoles a distance r apart: $\vec{p}_1$ points perpendicular to the line joining them, $\vec{p}_2$ points along that line. Find the torque on $\vec{p}_2$ due to $\vec{p}_1$, and the torque on $\vec{p}_1$ due to $\vec{p}_2$.

Solution.

[Diagram: $\vec{p}_1$ pointing up at the left; $\vec{p}_2$ a distance r to the right, pointing along the line joining them.]

$$\vec{E}_{dip} = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

Field of $\vec{p}_1$ @ $\vec{p}_2$: $\theta_{p_2} = \frac{\pi}{2}$. [Small diagram: $\vec{p}_1$ pointing up, the field point a distance r along the horizontal, angle $\pi/2$ marked.]

$$\vec{E}_1 = \frac{p_1}{4\pi\epsilon_0 r^3} \left(2 \cos \left(\frac{\pi}{2} \right) \hat{r} + \sin \left(\frac{\pi}{2} \right) \hat{\theta} \right)$$

(the first term vanishes)

$$\vec{E}_1 = \frac{p_1}{4\pi\epsilon_0} \frac{1}{r^3} \hat{\theta}$$

Torque on $\vec{p}_2$:

$$\vec{N}_2 = \vec{p}_2 \times \vec{E}_1 = p_2 E_1 \sin(90) = p_2 E_1$$

$$\vec{N}_2 = \frac{p_1 p_2}{4\pi\epsilon_0 r^3} \quad \text{into the page}$$

Field of $\vec{p}_2$ @ $\vec{p}_1$ ($\theta = \pi$): [Small diagram: the angle from $\vec{p}_2$'s axis back to $\vec{p}_1$ is π .]

$$\vec{E}_2 = \frac{p_2}{4\pi\epsilon_0 r^3} (-2\hat{r})$$

$$\vec{N}_1 = \frac{2p_1 p_2}{4\pi\epsilon_0 r^3}$$

P41. P 4.6 Dipole Method of Images

(p. 76)

A perfect dipole $\vec{p}$ sits a height z above an infinite grounded conducting plane, tilted an angle θ from the perpendicular; find the torque on $\vec{p}$ by replacing the plane with an image dipole a distance $2z$ away.

Solution.

[Diagram: dipole $\vec{p}$ at angle θ a height z above a hatched grounded plane; an arrow points to the image construction, with the mirrored image dipole (its $+$ and $-$ ends marked) below the plane.]

[Diagram: image dipole $\vec{p}_i$ a distance $r \equiv 2z$ below the real dipole $\vec{p}$; each makes angle θ with the line joining them.]

$$\vec{E}_{dip} = \frac{p}{4\pi\epsilon_0 (r^3)} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

$$\vec{p} = p (\cos \theta \hat{r} + \sin \theta \hat{\theta})$$

$$\vec{N} = \vec{p} \times \vec{E}_i = \frac{p^2}{4\pi\epsilon_0 (2z)^3} \left[(\cos \theta \hat{r} + \sin \theta \hat{\theta}) \times (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}) \right]$$

$$\vec{N} = \frac{p^2 \sin \theta \cos \theta}{4\pi\epsilon_0 (2z)^3} (-\hat{\phi}) \quad \text{out of page}$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$

$N = \frac{p^2 \sin 2\theta}{4\pi\epsilon_0 16z^3} \quad \text{out of page}$
--

$$0 < \theta < \frac{\pi}{2}: \text{C.C.W.} \quad \frac{\pi}{2} \rightarrow \pi: \text{C.W.}$$

P42. Prob 3.28 Multipole Expansion of Loop

(pp. 77-78)

Compute the dipole moment of the surface charge $\sigma = K \cos \theta$ on a sphere of radius R (the shell of Ex 3.9) and the resulting dipole potential. *[Header: a struck-out word is overwritten with "loop", and the title trails "w/" plus a cursive symbol read as z; the notes call the distribution a "loop" throughout, but the object worked is the spherical shell.]*

Solution.

$$\vec{p} \equiv \int \vec{r}' \rho(\vec{r}') d\tau' \quad \vec{p} = q\vec{d}$$

$\vec{E}$ of loop is $\hat{z}$ due to symmetry, so

$$\vec{p} = p\hat{z}$$

$$p = \int z \rho d\tau \quad p = \int z \sigma da$$

[Margin: $\rho \sim K \cos \theta$; $z = R \cos \theta$; $da = R^2 \sin \theta d\theta d\phi$.]

$$p = \int_0^\pi R \cos \theta K \cos \theta R^2 \sin \theta d\theta \int_0^{2\pi} d\phi$$

$$p = 2\pi R^3 K \int_0^\pi \cos^2 \theta \sin \theta d\theta$$

$$u \equiv \cos \theta \quad du = -\sin \theta d\theta$$

$$\int_0^\pi \cos^2 \theta \sin \theta d\theta = -\int u^2 du \rightarrow -\frac{u^3}{3}$$

$$\rightarrow -\frac{\cos^3 \theta}{3} \Big|_0^\pi = \frac{-(-1)^3}{3} + \frac{1}{3} = \frac{2}{3}$$

$$p = \frac{4}{3}\pi R^3 K \quad \boxed{\vec{p} = \frac{4\pi R^3 K}{3} \hat{z}}$$

$$V_{dip} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

Cancelling the 4π 's:

$$V_{dip} = \frac{1}{4\pi\epsilon_0} \frac{4\pi R^3}{3} \frac{K \cos \theta}{r^2} = \boxed{\frac{KR^3 \cos \theta}{3\epsilon_0 r^2}} = V_{dip}$$

[The final label is squeezed against the page edge; it matches the exact V_{out} found in Ex 3.9.]

P43. Prob 7.8

(p. 79)

A square loop of side a sits a distance s above a long straight wire carrying current I . (a) Find the flux of $\vec{B}$ through the loop; (b) find the EMF $\mathcal{E}$ generated by pulling the loop directly away from the wire; (c) find $\mathcal{E}$ if the loop instead moves to the right at speed v .

[Diagram: square loop of side a , its bottom edge a distance s above a long horizontal wire carrying current I to the right.]

Solution.

$$\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad \Phi = \int \vec{B} \cdot d\vec{a}$$

a)

$$\Phi = \frac{\mu_0 I}{2\pi} \int_s^{s+a} \frac{1}{s} a ds = \frac{\mu_0 I a}{2\pi} \ln \left(\frac{s+a}{s} \right)$$

b)

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{\mu_0 I a}{2\pi} \left(\frac{ds}{dt} \frac{1}{s+a} - \frac{1}{s} \frac{ds}{dt} \right)$$

$$\frac{ds}{dt} = v$$

$$\mathcal{E} = -\frac{\mu_0 I a v}{2\pi} \left(\frac{s - (s+a)}{s(s+a)} \right) = \frac{\mu_0 I a^2 v}{2\pi s(s+a)}$$

counterclockwise

c)

$$\mathcal{E} = 0 \quad \Phi = \text{constant}$$

P44. Proof: $\mathcal{E} = -\frac{d\Phi}{dt}$

(p. 80)

Proof of the flux rule for motional EMF: for a loop moving through a magnetic field, the EMF equals minus the rate of change of the flux. Here $\vec{v}$ is the loop's velocity, $\vec{u}$ the charge velocity along the wire, and $\vec{w} = \vec{v} + \vec{u}$ the total charge velocity.

Solution.

$$d\Phi = \Phi(t + dt) - \Phi(t) = \Phi_{\text{ribbon}} = \int_{\text{ribbon}} \vec{B} \cdot d\vec{a}$$

$$\vec{w} = \vec{v} + \vec{u}$$

$$d\vec{a} = (\vec{v} \times d\vec{l}) dt$$

$$\frac{d\Phi}{dt} = \oint \vec{B} \cdot (\vec{v} \times d\vec{l}) \quad (\vec{u} \text{ is } \parallel \text{ to } d\vec{l})$$

so

$$\frac{d\Phi}{dt} = \oint \vec{B} \cdot (\vec{w} \times d\vec{l})$$

Scalar triple-product can be rewritten

$$\vec{B} \cdot (\vec{w} \times d\vec{l}) = -(\vec{w} \times \vec{B}) \cdot d\vec{l}$$

$$\frac{d\Phi}{dt} = - \oint \underbrace{(\vec{w} \times \vec{B})}_{(\equiv \vec{f}_{\text{mag}})} \cdot d\vec{l} = - \oint \vec{f}_{\text{mag}} \cdot d\vec{l}$$

$$\boxed{\mathcal{E} = -\frac{d\Phi}{dt}}$$

P45. Electric Dipole Radiation

(pp. 81-84)

Derivation of the potentials, radiation-zone fields, and radiated power of an oscillating electric dipole: charges $\pm q(t) = \pm q_0 \cos(\omega t)$ separated by d on the z -axis, so $\vec{p}(t) = p_0 \cos(\omega t) \hat{z}$ with $p_0 \equiv q_0 d$, carried through the three standard approximations ($d \ll r$, $d \ll c/\omega$, $r \gg c/\omega$) to $\langle \vec{S} \rangle$ and $\langle P \rangle$.

Solution.

$$\vec{p}(t) = p_0 \cos(\omega t) \hat{z} \quad q(t) = q_0 \cos(\omega t) \quad p_0 \equiv q_0 d$$

$$V_{\text{ret}} = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_0 \cos[\omega(t - \mathbf{r}_+/c)]}{\mathbf{r}_+} - \frac{q_0 \cos[\omega(t - \mathbf{r}_-/c)]}{\mathbf{r}_-} \right\}$$

$$\mathbf{r}_{\pm} = \sqrt{r^2 \mp rd \cos \theta + \left(\frac{d}{2}\right)^2}$$

approx. 1: $d \ll r$

$$\mathbf{r}_{\pm} \approx r \left(1 \mp \frac{d}{2r} \cos \theta \right)$$

$$\frac{1}{\mathbf{r}_{\pm}} \approx \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right) \quad (1)$$

$$\begin{aligned} \cos[\omega(t - \mathbf{r}_{\pm}/c)] &\approx \cos \left[\omega(t - r/c) \pm \frac{\omega d}{2c} \cos \theta \right] \\ &= \cos[\omega(t - r/c)] \cos \left(\frac{\omega d}{2c} \cos \theta \right) \mp \sin[\omega(t - r/c)] \sin \left(\frac{\omega d}{2c} \cos \theta \right) \end{aligned}$$

approx 2: $d \ll \frac{c}{\omega}$, with $\lambda = \frac{2\pi c}{\omega}$, $\rightarrow d \ll \frac{\lambda}{2\pi}$

$$\cos[\omega(t - \mathbf{r}_{\pm}/c)] \approx \cos[\omega(t - r/c)] \mp \frac{\omega d}{2c} \cos \theta \sin[\omega(t - r/c)] \quad (2)$$

(1), (2) $\rightarrow V_{\text{ret}}$:

$$V = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left\{ -\frac{\omega}{c} \sin[\omega(t - r/c)] + \frac{1}{r} \cos[\omega(t - r/c)] \right\}$$

approx 3: ($r \gg \frac{c}{\omega}$), $r \gg \lambda$

$$V = -\frac{p_0\omega}{4\pi\epsilon_0 c} \left(\frac{\cos\theta}{r} \right) \sin[\omega(t-r/c)]$$

$$\vec{I}(t) = \frac{dq}{dt} \hat{z} = -q_0\omega \sin(\omega t) \hat{z}$$

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int_{-d/2}^{d/2} \frac{-q_0\omega \sin[\omega(t-\mathbf{r}/c)]}{\mathbf{r}} \hat{z} dz \quad (\text{margin: } p_0 = q_0 d)$$

$$\vec{A} = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin[\omega(t-r/c)] \hat{z}$$

$$\nabla V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} \approx \frac{p_0\omega^2}{4\pi\epsilon_0 c^2} \left(\frac{\cos\theta}{r} \right) \cos[\omega(t-r/c)] \hat{r}$$

$$\frac{\partial \vec{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi r} \cos[\omega(t-r/c)] \left(\cos\theta \hat{r} - \sin\theta \hat{\theta} \right)$$

$$\vec{E} = -\nabla V - \frac{\partial \vec{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi} \left(\frac{\sin\theta}{r} \right) \cos[\omega(t-r/c)] \hat{\theta}$$

$$\nabla \times \vec{A} = \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\phi} = -\frac{\mu_0 p_0 \omega}{4\pi r} \left\{ \frac{\omega}{c} \sin\theta \cos[\omega(t-r/c)] + \frac{\sin\theta}{r} \sin[\omega(t-r/c)] \right\} \hat{\phi}$$

$$\vec{B} = \nabla \times \vec{A} = -\frac{\mu_0 p_0 \omega^2}{4\pi c} \left(\frac{\sin\theta}{r} \right) \cos[\omega(t-r/c)] \hat{\phi}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{\mu_0}{c} \left\{ \frac{p_0\omega^2}{4\pi} \frac{\sin\theta}{r} \cos[\omega(t-r/c)] \right\}^2 \hat{r}$$

$$\langle \vec{S} \rangle = \left(\frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2\theta}{r^2} \hat{r}$$

$$\langle P \rangle = \int \langle \vec{S} \rangle \cdot d\vec{a} = \frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \int \frac{\sin^2\theta}{r^2} r^2 \sin\theta d\theta d\phi$$

$$\langle P \rangle = \frac{\mu_0 p_0^2 \omega^4}{12\pi c}$$

P46. Magnetic Dipole Radiation

(pp. 85-88)

The same program for an oscillating magnetic dipole: a wire loop of radius b in the xy -plane carrying $I(t) = I_0 \cos(\omega t)$, so $\vec{m}(t) = \pi b^2 I(t) \hat{z} = m_0 \cos(\omega t) \hat{z}$; the vector potential is expanded through the three approximations, then the radiation fields, $\langle \vec{S} \rangle$, $\langle P \rangle$, and the ratio $P_{\text{mag}}/P_{\text{ele}}$.

[Diagram: circular loop of radius b in the xy -plane; field point at $\vec{r}$ (polar angle θ from $\hat{z}$, drawn above the x -axis, i.e. in the xz -plane); source element $d\vec{l}'$ on the loop at azimuth ϕ' , separation r to the field point; ψ is the angle between $\vec{r}$ and $\vec{b}$.]

Solution.

$$I(t) = I_0 \cos(\omega t)$$

$$\vec{m}(t) = \pi b^2 I(t) \hat{z} = m_0 \cos(\omega t) \hat{z} \quad m_0 \equiv \pi b^2 I_0$$

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \frac{I_0 \cos[\omega(t - r/c)]}{r} d\vec{l}'$$

$\vec{r}$ is above $\hat{x}$ so $\vec{A} = A \hat{y}$ (x -axis cancels)

$$\vec{A}(\vec{r}, t) = \frac{\mu_0 I_0 b}{4\pi} \hat{y} \int_0^{2\pi} \frac{\cos[\omega(t - r/c)]}{r} \cos \phi' d\phi' \quad (1)$$

law of cosines

$$r = \sqrt{r^2 + b^2 - 2rb \cos \psi}$$

$$\vec{r} = r \sin \theta \hat{x} + r \cos \theta \hat{z}$$

$$\vec{b} = b \cos \phi' \hat{x} + b \sin \phi' \hat{y}$$

$$\vec{r} \cdot \vec{b} = rb \cos \psi = rb \sin \theta \cos \phi'$$

$$r = \sqrt{r^2 + b^2 - 2rb \sin \theta \cos \phi'}$$

approx 1: $b \ll r$

$$\begin{aligned}\mathfrak{r} &\approx r \left(1 - \frac{b}{r} \sin \theta \cos \phi'\right) \\ \frac{1}{\mathfrak{r}} &\approx \frac{1}{r} \left(1 + \frac{b}{r} \sin \theta \cos \phi'\right)\end{aligned}\tag{2}$$

$$\begin{aligned}\cos[\omega(t - \mathfrak{r}/c)] &\approx \cos \left[\omega(t - r/c) + \frac{\omega b}{c} \sin \theta \cos \phi' \right] \\ &= \cos[\omega(t - r/c)] \cos \left(\frac{\omega b}{c} \sin \theta \cos \phi' \right) - \sin[\omega(t - r/c)] \sin \left(\frac{\omega b}{c} \sin \theta \cos \phi' \right)\end{aligned}$$

approx 2: $b \ll \frac{c}{\omega}$

$$\cos[\omega(t - \mathfrak{r}/c)] \approx \cos[\omega(t - r/c)] - \frac{\omega b}{c} \sin \theta \cos \phi' \sin[\omega(t - r/c)]\tag{3}$$

(2), (3) $\rightarrow$ (1) to the 1st order

$$\vec{A}(\vec{r}, t) \approx \frac{\mu_0 I_0 b}{4\pi r} \hat{y} \int_0^{2\pi} \left\{ \cos[\omega(t - r/c)] + b \sin \theta \cos \phi' \left(\frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right) \right\} \cos \phi' d\phi'$$

1st term $\rightarrow 0$

$$\int_0^{2\pi} \cos \phi' d\phi' = 0$$

2nd term

$$\begin{aligned}\vec{A} &= \frac{\mu_0 m_0}{4\pi} \left(\frac{\sin \theta}{r} \right) \left\{ \frac{1}{r} \cos[\omega(t - r/c)] - \frac{\omega}{c} \sin[\omega(t - r/c)] \right\} \hat{\phi}\end{aligned}$$

Static ($\omega = 0$) gives mag. dipole

$$\vec{A} = \frac{\mu_0}{4\pi} \frac{m_0 \sin \theta}{r^2} \hat{\phi}$$

approx. 3: $(r \gg \frac{c}{\omega}) \rightarrow \frac{\omega}{c} \gg \frac{1}{r}$

$$\boxed{\vec{A} = -\frac{\mu_0 m_0 \omega}{4\pi c} \left(\frac{\sin \theta}{r} \right) \sin[\omega(t - r/c)] \hat{\phi}}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\phi}$$

$$\vec{B} = \nabla \times \vec{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\theta}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{\mu_0}{c} \left\{ \frac{m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \right\}^2 \hat{r}$$

$$\langle \vec{S} \rangle = \left(\frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \right) \frac{\sin^2 \theta}{r^2} \hat{r}$$

$$\langle P \rangle = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}$$

$$\frac{P_{\text{mag}}}{P_{\text{ele}}} = \left(\frac{m_0}{p_0 c} \right)^2 \quad m_0 = \pi b^2 I_0, \quad p_0 = q_0 d, \quad I_0 = q_0 \omega, \quad d = \pi b$$

$$\frac{P_{\text{mag}}}{P_{\text{ele}}} = \left(\frac{\omega b}{c} \right)^2$$

P47. Prob 5.28 — Coulomb Gauge

(pp. 89-91)

For the magnetostatic vector potential $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{\mathbf{r}} d\tau'$, show the three Coulomb-gauge properties: $\nabla \cdot \vec{A} = 0$, $\nabla \times \vec{A} = \vec{B}$, and $\nabla^2 \vec{A} = -\mu_0 \vec{J}$, where $\vec{B}$ is the Biot-Savart field.

Solution.

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{\mathbf{r}} d\tau'$$

$$\nabla \cdot \vec{A} = 0 \quad \nabla \times \vec{A} = \vec{B} \quad \nabla^2 \vec{A} = -\mu_0 \vec{J} \quad \text{with} \quad \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}') \times \hat{\mathbf{r}}}{\mathbf{r}^2} d\tau'$$

$$\nabla \cdot \vec{A} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau'$$

$$\nabla \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) = \frac{1}{\mathbf{r}} \left(\cancel{\nabla \cdot \vec{J}} \right) + \vec{J} \cdot \nabla \left(\frac{1}{\mathbf{r}} \right)$$

$$\nabla \left(\frac{1}{\mathbf{r}} \right) = -\nabla' \frac{1}{\mathbf{r}} \quad \vec{\mathbf{r}} = \vec{r} - \vec{r}'$$

$$\nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) = \frac{1}{\mathbf{r}} \left(\cancel{\nabla' \cdot \vec{J}} \right) + \vec{J} \cdot \nabla' \left(\frac{1}{\mathbf{r}} \right)$$

$$\nabla' \cdot \vec{J} = 0 \quad (\text{magnetostatics})$$

$$\nabla \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) = -\nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right)$$

$$\nabla \cdot \vec{A} = -\frac{\mu_0}{4\pi} \int \nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau' \xrightarrow{\text{div. theorem}} -\frac{\mu_0}{4\pi} \oint \frac{\vec{J}}{\mathbf{r}} \cdot d\vec{a}'$$

where $\vec{J} = 0$ on surface $\therefore \nabla \cdot \vec{A} = 0$

$$\nabla \times \vec{A} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau' = \frac{\mu_0}{4\pi} \int \left[\frac{1}{\mathbf{r}} \left(\cancel{\nabla \times \vec{J}} \right) - \vec{J} \times \nabla \left(\frac{1}{\mathbf{r}} \right) \right] d\tau'$$

$$\vec{J} \neq f(\vec{r})$$

$$\nabla \left(\frac{1}{\mathbf{r}} \right) = -\frac{\vec{\mathbf{r}}}{\mathbf{r}^3} = -\frac{\hat{\mathbf{t}}}{\mathbf{r}^2}$$

$$\nabla \times \vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{J} \times \hat{\mathbf{t}}}{\mathbf{r}^2} d\tau' = \vec{B}$$

$$\nabla^2 \vec{A} = \frac{\mu_0}{4\pi} \int \nabla^2 \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau'$$

$$\nabla^2 f = \nabla \cdot (\nabla f) \quad \text{where } \nabla \cdot \vec{J} = 0$$

so $\nabla^2 \vec{J} = 0$, so then

$$\nabla^2 \left(\frac{\vec{J}}{\mathbf{r}} \right) = \vec{J} \nabla^2 \left(\frac{1}{\mathbf{r}} \right) \quad \text{margin: } \nabla \left(\frac{1}{\mathbf{r}} \right) = -\frac{\hat{\mathbf{t}}}{\mathbf{r}^2}, \quad \nabla \cdot \left(\frac{\vec{\mathbf{r}}}{\mathbf{r}^3} \right) = 4\pi \delta(\vec{\mathbf{r}})$$

so

$$\nabla^2 \left(\frac{1}{\mathbf{r}} \right) = -4\pi \delta(\vec{\mathbf{r}})$$

$$\nabla^2 \vec{A} = -\frac{\mu_0}{4\pi} \int \vec{J}(\vec{r}') 4\pi \delta(\vec{\mathbf{r}}) d\tau' \quad \delta(\vec{\mathbf{r}}) = \delta(\vec{r} - \vec{r}')$$

thus

$$\boxed{\nabla^2 \vec{A} = -\mu_0 \vec{J}(\vec{r})}$$

P48. Prob 10.10 — Lorenz Gauge

(pp. 92-94)

Show that the retarded potentials satisfy the Lorenz gauge condition $\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \partial V / \partial t$. The work turns on relating $\nabla \cdot \vec{J}$ and $\nabla' \cdot \vec{J}$ for the retarded current $\vec{J}(\vec{r}', t_r)$, whose dependence on $\vec{r}$ and $\vec{r}'$ also enters through $t_r = t - \mathfrak{r}/c$.

Solution.

$$\nabla \cdot \left(\frac{\vec{J}}{\mathfrak{r}} \right) = \frac{1}{\mathfrak{r}} (\nabla \cdot \vec{J}) + \vec{J} \cdot \nabla \left(\frac{1}{\mathfrak{r}} \right)$$

$$\nabla' \cdot \left(\frac{\vec{J}}{\mathfrak{r}} \right) = \frac{1}{\mathfrak{r}} (\nabla' \cdot \vec{J}) + \vec{J} \cdot \left(\nabla' \frac{1}{\mathfrak{r}} \right)$$

$$\nabla \left(\frac{1}{\mathfrak{r}} \right) = -\nabla' \left(\frac{1}{\mathfrak{r}} \right) \quad \vec{\mathfrak{r}} = \vec{r} - \vec{r}'$$

$$\begin{aligned} \nabla \cdot \left(\frac{\vec{J}}{\mathfrak{r}} \right) &= \frac{1}{\mathfrak{r}} (\nabla \cdot \vec{J}) - \vec{J} \cdot \nabla' \left(\frac{1}{\mathfrak{r}} \right) \\ &= \frac{1}{\mathfrak{r}} (\nabla \cdot \vec{J}) + \frac{1}{\mathfrak{r}} (\nabla' \cdot \vec{J}) - \nabla' \cdot \frac{\vec{J}}{\mathfrak{r}} \end{aligned}$$

$$\nabla \cdot \vec{J} = \partial_i J_i = \partial_{t_r} \frac{\partial t_r}{\partial x^i} J_i$$

$$\frac{\partial t_r}{\partial x^i} = -\frac{1}{c} \frac{\partial \mathfrak{r}}{\partial x^i} = -\frac{1}{c} \nabla \mathfrak{r}$$

$$\nabla \cdot \vec{J} = -\frac{1}{c} \partial_{t_r} \vec{J} \cdot (\nabla \mathfrak{r}) = -\frac{1}{c} \frac{\partial \vec{J}}{\partial t_r} \cdot (\nabla \mathfrak{r})$$

Similarly

$$\nabla' \cdot \vec{J} = -\frac{\partial \rho}{\partial t} - \frac{1}{c} \frac{\partial \vec{J}}{\partial t_r} \cdot (\nabla' \mathfrak{r}) \quad \left(\nabla' \cdot \vec{J}(\vec{r}') = -\frac{\partial \rho}{\partial t} \right)$$

$$\begin{aligned} \nabla \cdot \left(\frac{\vec{J}}{\mathfrak{r}} \right) &= \frac{1}{\mathfrak{r}} (\nabla \cdot \vec{J}) + \frac{1}{\mathfrak{r}} (\nabla' \cdot \vec{J}) - \nabla' \cdot \left(\frac{\vec{J}}{\mathfrak{r}} \right) \\ &= \frac{1}{\mathfrak{r}} \left[\cancel{-\frac{1}{c} \frac{\partial \vec{J}}{\partial t_r} \cdot (\nabla \mathfrak{r})} - \frac{\partial \rho}{\partial t} + \cancel{\frac{1}{c} \frac{\partial \vec{J}}{\partial t_r} \cdot (\nabla \mathfrak{r})} \right] - \nabla' \cdot \frac{\vec{J}}{\mathfrak{r}} \end{aligned}$$

$$\nabla \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) = -\frac{1}{\mathbf{r}} \frac{\partial \rho}{\partial t} - \nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right)$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{\mathbf{r}} d\tau'$$

$$\nabla \cdot \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \left(\nabla \cdot \frac{\vec{J}}{\mathbf{r}} \right) d\tau'$$

$$\begin{aligned} \nabla \cdot \vec{A} &= \frac{\mu_0}{4\pi} \left[\frac{\partial}{\partial t} \int -\frac{\rho}{\mathbf{r}} d\tau - \int \nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau \right] \\ &= -\frac{\mu_0}{4\pi} \left[\frac{\partial}{\partial t} \int \frac{\rho}{\mathbf{r}} d\tau + \int \nabla' \cdot \left(\frac{\vec{J}}{\mathbf{r}} \right) d\tau \right] \end{aligned}$$

$$\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \left[\frac{1}{4\pi \epsilon_0} \int \frac{\rho}{\mathbf{r}} d\tau \right] - \cancel{\frac{\mu_0}{4\pi} \oint \frac{\vec{J}}{\mathbf{r}} \cdot d\vec{a}} \quad (\text{div. theorem; where } \vec{J} = 0 \text{ on surface})$$

$$\nabla \cdot \vec{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

P49. Ex 2.6 — Two Infinite Planes

(p. 95)

Two infinite parallel planes carry equal but opposite uniform charge densities $\pm\sigma$; find $\vec{E}$ on the left, middle, and right. The page argues entirely by superposition diagrams (field directions of each sheet in the three regions); no formulas are written.

Solution.

2 infinite planes (parallel) equal but opposite uniform charge densities $\pm\sigma$. Find: $\vec{E}$ on the left, middle, right.

[Diagram: two parallel rectangular sheets drawn side by side, labeled $+\sigma$ and $-\sigma$.]

[Diagram: the planes as two vertical lines ($+\sigma$ left, $-\sigma$ right) dividing space into three regions. Top row, the $+$ sheet's field E_+ : points left in the left region, right in the middle region, right in the right region (always away from $+\sigma$). Bottom row, the $-$ sheet's field E_- : points right in the left region, right in the middle region, left in the right region (always toward $-\sigma$). Outside the pair the two arrows oppose; between the planes they point the same way.]

[Diagram: between the two planes, arrows labeled $\vec{E}$ pointing from $+\sigma$ toward $-\sigma$ — the total field.]

so

[Diagram: the gap again, planes labeled σ_+ and σ_- , with the two contributions E_+ and E_- drawn parallel, both pointing from σ_+ to σ_- .]

where if so, then by attraction

[Diagram: at the σ_+ plane, E_+ points away to the left while E_- points right, toward the $-$ plane; at the σ_- plane, E_+ points right (away from $+\sigma$) while E_- points left, into it — each sheet sits in the other's field, pulling the sheets together.]

& Newton Law (3rd) *[crossed-out word, illegible]*

P50. Spherical Laplace (Legendre, Rodrigues)

(pp. 96–97)

[reference notes, not a worked problem]

General solution of Laplace's equation in spherical coordinates with azimuthal symmetry (margin label "3D Potential"): separation of variables $V = R(r)\Theta(\theta)$, the radial and angular equations with separation constant $\ell(\ell+1)$, Legendre polynomials via the Rodrigues formula, and the general series for $V(r, \theta)$.

Solution.

$$d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

$$d\tau = r^2 \sin \theta dr d\theta d\phi$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

Azimuthal symmetry (ind. of ϕ)

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0$$

Set $V(r, \theta) = R(r) \Theta(\theta)$, then divide by $R(r) \Theta(\theta)$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0$$

for convenience

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell+1) \quad \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\ell(\ell+1)$$

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell+1) R$$

$$R(r) = A r^\ell + \frac{B}{r^{\ell+1}}$$

$$\frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\ell(\ell+1) \sin \theta \Theta$$

Legendre Polynomials

$$\Theta(\theta) = P_\ell(\cos \theta)$$

Rodrigues formula

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell$$

$$P_0(x) = 1 \quad P_1(x) = x \quad P_2(x) = \frac{3x^2 - 1}{2} \quad P_3 = \frac{5x^3 - 3x}{2} \quad P_4 = \frac{35x^4 - 30x^2 + 3}{8} \quad \text{etc.} \dots$$

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta)$$

P51. Infinite Pipe 2D

(pp. 98-100)

Laplace's equation is solved for the potential inside an infinitely long rectangular pipe (axis along z) of cross-section $-b \leq x \leq b$, $0 \leq y \leq a$: the walls $y = 0$ and $y = a$ are grounded, and the two side walls at $x = \pm b$ are held at constant potential V_0 .

Solution.

[Diagram: a rectangular pipe extending to infinity along z (jagged ends); the faces $y = 0$ and $y = a$ are marked $V = 0$, the side faces at $x = -b$ and $x = +b$ are marked V_0 .]

$$V(x, 0) = V(x, a) = 0, \quad V(\pm b, y) = V_0$$

$$\{x \mid x \in \pm b\}, \quad \{y \mid y \in 0, a\}$$

$$\nabla^2 V = 0 \rightarrow \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0$$

Again [a crossed-out symbol] has trivial solutions so

$$X \sim f(e^{\pm kx}) \quad Y \sim f(\sin + \cos)$$

X is symmetric, $\{x \mid x \in \pm b\}$:

$$e^{kx} + e^{-kx} = 2 \cosh(kx)$$

$$V(x, y) = \cosh(kx) (C \sin ky + D \cos ky)$$

$$V(x, 0) = 0 = \cosh(kx) (C \sin(0) + D \cos(0))$$

so $D \rightarrow 0$.

$$V(x, y) = C \cosh(kx) \sin(ky)$$

$$ka = n\pi \rightarrow k = \frac{n\pi}{a}$$

$$V(x, y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi x}{a}\right) \sin\left(\frac{n\pi y}{a}\right)$$

$$V(b, y) = \sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi b}{a}\right) \sin\left(\frac{n\pi y}{a}\right) = V_0$$

$$\sum_{n=1}^{\infty} C_n \int_0^a \cosh\left(\frac{n\pi b}{a}\right) \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = V_0 \int_0^a \sin\left(\frac{n'\pi y}{a}\right) dy$$

$$\sum_{n=1}^{\infty} C_n \cosh\left(\frac{n\pi b}{a}\right) \underbrace{\sin\left(\frac{n\pi y}{a}\right)}_{f(y)} = V_0$$

$$C_n = \frac{2V_0}{a \cosh\left(\frac{n\pi b}{a}\right)} \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy, \quad \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy = \frac{2a}{n\pi} \text{ if } n \text{ is odd}$$

$$C_n = \frac{4V_0}{n\pi} \frac{1}{\cosh\left(\frac{n\pi b}{a}\right)}$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} \frac{\cosh\left(\frac{n\pi x}{a}\right)}{\cosh\left(\frac{n\pi b}{a}\right)} \sin\left(\frac{n\pi y}{a}\right)$$

P52. Inf. Rect. Pipe 3D

(pp. 101-103)

Laplace's equation is solved in the semi-infinite rectangular pipe $\{x \mid x \in [0, \infty]\}$, $\{y \mid y \in [0, a]\}$, $\{z \mid z \in [0, b]\}$: the four long walls are grounded, $V \rightarrow 0$ as $x \rightarrow \infty$, and the end face at $x = 0$ is held at a prescribed potential $V_0(y, z)$.

Solution.

$$V(x, 0, z) = V(x, a, z) = 0, \quad V(x, y, 0) = V(x, y, b) = 0$$

$$V \rightarrow 0 : V(\infty, y, z), \quad V(0, y, z) = V_0(y, z)$$

[Diagram: semi-infinite rectangular duct along +x with jagged open far end; height a along y, depth b along z; the x = 0 face carries V₀(y, z), the bottom face is marked V = 0, and V → 0 toward the far end.]

Laplace eq.

$$\nabla^2 V = 0, \quad \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$V(x, y, z) = X(x) Y(y) Z(z)$$

$$\frac{d^2 X}{dx^2} Y Z + \frac{d^2 Y}{dy^2} X Z + X Y \frac{d^2 Z}{dz^2} = 0$$

y & z have set boundaries a, b so sin & cos:

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -k^2 \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = -\ell^2$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = k^2 + \ell^2$$

$$X(x) = A e^{\sqrt{k^2 + \ell^2} x} + B e^{-\sqrt{k^2 + \ell^2} x}$$

$$Y(y) = C \sin(ky) + D \cos(ky)$$

$$Z(z) = E \sin(\ell z) + F \cos(\ell z)$$

$$V(y = 0) \rightarrow D = 0, \quad V(z = 0) \rightarrow F = 0, \quad V(x \rightarrow \infty) \rightarrow A = 0$$

let $BCE \equiv C$:

$$V = C e^{-\sqrt{k^2 + \ell^2} x} \sin(ky) \sin(\ell z)$$

$$ka = n\pi \rightarrow k = \frac{n\pi}{a}, \quad \ell = \frac{m\pi}{b}$$

$$V = \sum_{n,m=1}^{\infty} C_{nm} e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi z}{b}\right)$$

$$V(x=0) = V_0(y, z) = \sum_{n,m=1}^{\infty} C_{nm} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi z}{b}\right)$$

$$\sum \sum C_{nm} \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy \int_0^b \sin\left(\frac{m\pi z}{b}\right) \sin\left(\frac{m'\pi z}{b}\right) dz = \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy \int_0^b \sin\left(\frac{m\pi z}{b}\right) dz V_0$$

Fourier's trick:

$$C_{nm} = \frac{2}{a} \frac{2}{b} \int_0^a \int_0^b V_0 \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi z}{b}\right) dy dz$$

$$C_{nm} = \frac{16 V_0}{nm \pi^2} \quad \text{if } n, m \text{ are odd}$$

(General form noted alongside, "fourier's trick":

$$C_{nm} = \frac{2}{L_y} \frac{2}{L_z} \int_0^{L_y} dy f_y(n') \int_0^{L_z} dz f_z(m') V_0$$

)

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{n,m=1,3,5\dots}^{\infty} e^{-\pi \sqrt{\left(\frac{n}{a}\right)^2 + \left(\frac{m}{b}\right)^2} x} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi z}{b}\right) \frac{1}{nm}$$

P53. Ex 3.2 — Sep. of Variables

(pp. 104-107)

The semi-infinite slot: two grounded semi-infinite plates at $y = 0$ and $y = a$ run from $x = 0$ to $x \rightarrow \infty$, closed at $x = 0$ by a strip held at $V_0(y)$, with $V \rightarrow 0$ as $x \rightarrow \infty$; $V(x, y)$ is found by separation of variables, and the coefficients are then evaluated for constant V_0 .

Solution.

[Diagram: two semi-infinite grounded plates ($V = 0$) parallel to the x - z plane at $y = 0$ and $y = a$, jagged at large x ; the strip $V_0(y)$ at $x = 0$ closes the slot; axes x, y, z drawn.]

$$V(x, 0) = V(x, a) = 0$$

$$V(0, y) = V_0(y), \quad V \rightarrow 0 \text{ as } x \rightarrow \infty$$

① Laplace's 2D eq.

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

②

$$V(x, y) = X(x)Y(y)$$

② $\rightarrow$ ①

$$\frac{d^2 X}{dx^2} Y + X \frac{d^2 Y}{dy^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0, \quad k^2 - k^2 = 0$$

with the first term marked $\equiv k^2$ and the second $-k^2$:

$$X(x) = A e^{kx} + B e^{-kx} \quad Y(y) = C \sin(ky) + D \cos(ky)$$

$$V(x, y) = \left(A e^{kx} + B e^{-kx} \right) (C \sin(ky) + D \cos(ky))$$

$$V(\infty, y) = \left(A e^\infty + \frac{B}{e^\infty} \right) Y(y) = 0$$

So A must go to zero. $A = 0$.

$$X(x) = B e^{-kx}$$

$$V(x, 0) = B e^{-kx} (C \sin(0) + D \cos(0)) = 0$$

so $D \rightarrow 0$.

$$V(x, y) = BC e^{-kx} \sin(ky) \quad \text{let } BC \equiv B$$

$$V(x, a) = 0 = B e^{-kx} \sin(ka)$$

$\sin(ka) = 0$ when $ka = n\pi$ so

$$k = \frac{n\pi}{a}$$

$$V(x, y) = \sum_{n=1}^{\infty} e^{-\frac{n\pi x}{a}} \sin\left(\frac{n\pi y}{a}\right) B_n$$

$$V(0, y) \rightarrow \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi y}{a}\right) = V_0(y)$$

integrate over y -axis with additional $f(n')$:

$$\sum_{n=1}^{\infty} B_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \int_0^a V_0(y) \sin\left(\frac{n'\pi y}{a}\right) dy$$

$$\int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \begin{cases} 0 & n' \neq n \\ \frac{a}{2} & n = n' \end{cases}$$

$$B_n \frac{a}{2} = \int_0^a V_0(y) \sin\left(\frac{n\pi y}{a}\right) dy$$

$$B_n = \frac{2V_0}{a} \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy = \frac{2V_0}{n\pi} (1 - \cos n\pi)$$

$$B_n = 0 \quad \text{if } n = \text{even}$$

so

$$B_n = \frac{4V_0}{n\pi} \quad \text{for odd } n$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5}^{\infty} \frac{1}{n} e^{-\frac{n\pi x}{a}} \sin\left(\frac{n\pi y}{a}\right)$$

P54. Spherical Coordinates → Multipole Expansion

(p. 108)

[reference notes, not a worked problem]

The spherical-harmonic expansion of $1/|\vec{x} - \vec{x}'|$ and the resulting multipole form of the potential of a charge density $\rho(\vec{x}')$, evaluated outside the sphere enclosing the charge ($r_{<} = r'$, $r_{>} = r$).

Solution.

$$\frac{1}{|\vec{x} - \vec{x}'|} = 4\pi \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{2\ell+1} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} Y_{\ell m}^*(\theta', \phi') Y_{\ell m}(\theta, \phi)$$

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

for outside the sphere $r_{<} = r'$, $r_{>} = r$:

$$\Phi(\vec{x}) = \frac{1}{\epsilon_0} \sum_{\ell, m} \frac{1}{2\ell+1} \int \frac{r'^{\ell}}{r^{\ell+1}} Y_{\ell m}^*(\theta', \phi') \rho(\vec{x}') Y_{\ell m}(\theta, \phi) d^3x'$$

P55. Eigenfunction Expansions for Green functions

(pp. 109-110)

[reference notes, not a worked problem]

The eigenfunction expansion of the Green function of the operator $\nabla^2 + f(\vec{x}) + \lambda$ is derived, then specialized to $f = 0$, $\lambda = 0$ with the free-space wave-equation continuum $\psi_{\vec{k}}$ (giving the Fourier representation of $1/|\vec{x} - \vec{x}'|$) and to the Dirichlet modes of a rectangular box $x \in (0, a)$, $y \in (0, b)$, $z \in (0, c)$.

Solution.

$$\nabla^2 \psi_n(\vec{x}) + (f(\vec{x}) + \lambda_n) \psi_n(\vec{x}) = 0$$

$$\int_V \psi_m^*(\vec{x}) \psi_n(\vec{x}) d^3x = \delta_{mn}$$

$$\nabla^2 G(\vec{x}, \vec{x}') + [f(\vec{x}) + \lambda] G(\vec{x}, \vec{x}') = -4\pi \delta(\vec{x} - \vec{x}')$$

$$G(\vec{x}, \vec{x}') = \sum_n a_n(\vec{x}') \psi_n(\vec{x})$$

$$\sum_m a_m(\vec{x}') (\lambda - \lambda_m) \psi_m(\vec{x}) = -4\pi \delta(\vec{x} - \vec{x}')$$

$\times \psi_n^*(\vec{x})$ (both sides) & int $\int_V$ — orthogonality simplifies LHS

$$\rightarrow a_n(\vec{x}') = 4\pi \frac{\psi_n^*(\vec{x}')}{\lambda_n - \lambda}$$

so

$$G(\vec{x}, \vec{x}') = 4\pi \sum_n \frac{\psi_n^*(\vec{x}') \psi_n(\vec{x})}{\lambda_n - \lambda}$$

let $f(\vec{x}) = 0$, $\lambda = 0 \rightarrow$ wave eq. over all space:

$$(\nabla^2 + k^2) \psi_{\vec{k}}(\vec{x}) = 0$$

$$\psi_{\vec{k}}(\vec{x}) = \frac{1}{(2\pi)^{3/2}} e^{i\vec{k} \cdot \vec{x}}$$

normalization δ -fun:

$$\int \psi_{\vec{k}'}^*(\vec{x}) \psi_{\vec{k}}(\vec{x}) d^3x = \delta(\vec{k} - \vec{k}')$$

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{1}{2\pi^2} \int d^3k \frac{e^{i\vec{k} \cdot (\vec{x} - \vec{x}')}}{k^2}$$

For Cartesian box $\{x \in (0, a); y \in (0, b); z \in (0, c)\}$:

$$(\nabla^2 + k_{lmn}^2) \psi_{lmn}(x, y, z) = 0$$

$$\psi_{lmn}(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{n\pi z}{c}\right)$$

$$k_{lmn}^2 = \pi^2 \left(\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2} \right)$$

$$G(\vec{x}, \vec{x}') = \frac{32}{\pi abc} \sum_{l,m,n=1}^{\infty} \frac{\sin\left(\frac{l\pi x}{a}\right) \sin\left(\frac{l\pi x'}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{m\pi y'}{b}\right) \sin\left(\frac{n\pi z}{c}\right) \sin\left(\frac{n\pi z'}{c}\right)}{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}}$$

more on pg 129 (*a textbook page reference*).

P56. Cylindrical Green's Functions

(pp. 111-115)

The free-space Green function is expanded in cylindrical coordinates (ρ, ϕ, z) : the δ -functions are written in Fourier form, the radial equation is solved with modified Bessel functions I_m, K_m normalized through the Wronskian/slope-discontinuity condition, the resulting expansion of $1/|\vec{x}-\vec{x}'|$ is used to obtain (and check) the K_0 addition theorem, and the $k \rightarrow 0$ limit gives the 2-D logarithmic expansion.

[p. 112 is a second scan of the same "1/4" page as p. 111; the entry's pages are corner-marked 1/4-4/4.]

Solution.

$$\nabla^2 G(\vec{x}, \vec{x}') = -\frac{4\pi}{\rho} \delta(\rho - \rho') \delta(\phi - \phi') \delta(z - z')$$

$$\delta(z - z') = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ik(z-z')} = \frac{1}{\pi} \int_0^{\infty} dk \cos[k(z - z')]$$

$$\delta(\phi - \phi') = \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} e^{im(\phi-\phi')}$$

$$G(\vec{x}, \vec{x}') = \frac{1}{2\pi^2} \sum_{m=-\infty}^{\infty} \int_0^{\infty} dk e^{im(\phi-\phi')} \cos[k(z - z')] g_m(k, \rho, \rho')$$

$$\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dg_m}{d\rho} \right) - \left(k^2 + \frac{m^2}{\rho^2} \right) g_m = -\frac{4\pi}{\rho} \delta(\rho - \rho')$$

Linear combinations of modified Bessel functions $I_m(k\rho), K_m(k\rho)$: $\psi_1(k\rho)$ satisfies $\rho < \rho'$; $\psi_2(k\rho)$ lin. ind., satisf. $\rho > \rho'$.

$$g_m(k, \rho, \rho') = \psi_1(k\rho_{<}) \psi_2(k\rho_{>})$$

Normalize wrt. discontinuity by δ -function:

$$\left. \frac{dg_m}{d\rho} \right|_+ - \left. \frac{dg_m}{d\rho} \right|_- = -\frac{4\pi}{\rho'}$$

$$\left. \frac{dg_m}{d\rho} \right|_+ - \left. \frac{dg_m}{d\rho} \right|_- = k (\psi_1 \psi_2' - \psi_2 \psi_1') = k W[\psi_1, \psi_2]$$

Sturm-Liouville type (radial Green function), so then

$$W[\psi_1(x), \psi_2(x)] = -\frac{4\pi}{x}$$

g_m must be finite at $\rho = 0$; vanish $\rho \rightarrow \infty$, so

$$\psi_1(k\rho) = A I_m(k\rho) \quad \psi_2(k\rho) = K_m(k\rho)$$

choosing $A = 4\pi$:

$$W[I_m(x), K_m(x)] = -\frac{1}{x}$$

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{2}{\pi} \sum_{m=-\infty}^{\infty} \int_0^{\infty} dk e^{im(\phi-\phi')} \cos[k(z-z')] I_m(k\rho_{<}) K_m(k\rho_{>})$$

in purely real terms

$$\frac{1}{|\vec{x} - \vec{x}'|} = \frac{4}{\pi} \int_0^{\infty} dk \cos[k(z-z')] \left\{ \frac{1}{2} I_0(k\rho_{<}) K_0(k\rho_{>}) + \sum_{m=1}^{\infty} \cos[m(\phi-\phi')] I_m(k\rho_{<}) K_m(k\rho_{>}) \right\} \quad (1)$$

(the braced quantity is labelled β). If $\vec{x}' \rightarrow 0$ only $m = 0$ survives:

$$\frac{1}{\sqrt{\rho^2 + z^2}} = \frac{2}{\pi} \int_0^{\infty} \cos(kz) K_0(k\rho) dk \quad (2)$$

(margin note: comparable to $1/|\vec{x}|$). let

$$\rho^2 \rightarrow R^2 = \rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi'), \quad R = \sqrt{\rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')}$$

eval. (1) @ $z' = 0$; RHS of (1) $\equiv$ (2) RHS:

$$K_0\left(k\sqrt{\rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')}\right) = I_0(k\rho_{<}) K_0(k\rho_{>}) + 2 \sum_{m=1}^{\infty} \cos[m(\phi - \phi')] I_m(k\rho_{<}) K_m(k\rho_{>})$$

checking — the common factor $\frac{2}{\pi} \int_0^{\infty} dk \cos(kz)$ is struck from both sides:

$$2 \frac{4}{\pi} \int_0^{\infty} dk \cos(kz) \{\beta\} = \frac{2}{\pi} \int_0^{\infty} \cos(kz) K_0(k\rho) dk$$

$$K_0[k(\rho = R)] = 2\beta \quad \checkmark$$

take $k \rightarrow 0$ (the addition theorem is rewritten at the top of the last page):

$$\lim_{k \rightarrow 0} K_0(k\rho) \Rightarrow \ln\left(\frac{1}{R^2}\right) \Big|_{\rho=R}$$

$$\lim_{k \rightarrow 0} I_0(k\rho_{<}) \rightarrow 2$$

[sic: as literal limits $I_0 \rightarrow 1$ and $K_0(x) \sim \ln(2/x) - \gamma$; the factors of 2 in these two lines are bookkeeping, distributed so that the final expansion below comes out correctly]

$$\lim_{k \rightarrow 0} K_m(k\rho_{>}) \rightarrow \left(\frac{1}{\rho_{>}}\right)^m \quad \lim_{k \rightarrow 0} K_0(k\rho_{>}) = \ln\left(\frac{1}{\rho_{>}}\right)$$

$$\lim_{k \rightarrow 0} I_m(k\rho_{<}) \rightarrow \frac{1}{m} (\rho_{<})^m$$

$$\ln\left(\frac{1}{\rho^2 + \rho'^2 - 2\rho\rho' \cos(\phi - \phi')}\right) = 2 \ln\left(\frac{1}{\rho_{>}}\right) + 2 \sum_{m=1}^{\infty} \frac{1}{m} \left(\frac{\rho_{<}}{\rho_{>}}\right)^m \cos[m(\phi - \phi')]$$

look at problem 2.17